Table 13-3 Descriptive Terms Used in the Development of a Quality of Life Scale in Dogs with Chronic Pain

Domains	Negative Descriptors	Positive Descriptors
Activity	Apathetic, apprehensive, lackluster, lethargic, listless, reluctant, sleepy, slowed, sluggish, tired weary	Active, boisterous, bouncy, energetic, lively, playful, tireless
Comfort	Complaining, groaning, moaning, pained, sore, stoic, uncomfortable	Comfortable, stretching
Appetite	Off food, picky (with regard to food)	Enthusiastic about food, greedy, interested in food, thirsty
Extroversion-introversion	Detached, quiet, subdued, unresponsive, unsociable, withdrawn	Affectionate, bold, curious, eager, excitable, friendly, fun loving, nosy, outgoing, sociable
Aggression	Aggressive, grumpy, irritable, territorial or protective	Good-natured, even tempered, placid
Anxiety	Anxious, cautious, distressed, frightened, nervous, panicky, strained, uneasy, upset	Accepting, easygoing, laid-back
Alertness	Depressed, dull, confused, uninterested	Alert, bright, inquisitive, interested, keen, obedient
Dependence	Attention seeking, clingy, comfort seeking, pathetic or pitiful	Confident, independent
Contentment	Miserable, sad, sorrowful, resigned, unhappy	Contented, happy
Consistency	Inconsistent	Consistent
Agitation	Agitated, crying, disturbed, panting, restless, unsettled, whining	Calm, at ease
Posture-mobility	Awkward, limping, stiff	Athletic, fit, relaxed
Compulsion	Compulsive	No terms

From Wiseman-Orr ML, Nolan AM, Reid J, Scott EM: Development of a questionnaire to measure the effects of chronic pain on health-related quality of life in dogs. *Am J Vet Res* 65;1077-1084, 2004.

determine the quality of life for dogs with chronic pain.[79] These types of measures may guide decision making and treatment interventions (Table 13-3).

Summary

Assessment of patients is important to document the benefits of a rehabilitation program and help to improve protocols. Assessment techniques may be relatively simple and inexpensive, or more elaborate with expensive equipment. New assessment techniques must be developed to allow adequate comparison of a patient's progress over time and make specific changes in protocols to achieve the best possible outcome.

REFERENCES

1. Yeung TS, Stratford PW, MacDermid JC: The timed up and go test for use on an inpatient orthopaedic rehabilitation ward, *J Orthop Sports Phys Ther* 38:410-417, 2008.
2. Hyytiäinen HK, Mölsä SH, Junnila JT et al: Use of bathroom scales in measuring asymmetry of hindlimb static weight bearing in dogs with osteoarthritis, *Vet Comp Orthop Traumatol* 25:390-396, 2012.
3. Phelps HA, Ramos V, Shires PK et al: The effect of measurement method on static weight distribution to all legs in dogs using the Quadruped Biofeedback System, *Vet Comp Orthop Traumatol* 20:108-112, 2007.
4. Hicks DA, Millis DL, Arnold GA et al: *Comparison of weightbearing at a stance vs trotting in dogs with rear limb lameness*. Proc 32nd Annual Conference of the Veterinary Orhtopedic Society, 2005.
5. Millis DL, Westling M, Westling W: *Comparison of weightbearing at a stance vs trotting in dogs with lameness*. Proc 7th International Association of Veterinary Rehabilitation and Physical Therapy Meeting. Vienna, Austria, August 2013.
6. Quinn MM, Keuler NS, Lu Y et al: Evaluation of agreement between numerical rating scales, visual analogue scoring scales, and force plate gait analysis in dogs, *Vet Surg* 36:360-367, 2007.
7. Waxman AS, Robinson DA, Evans RB et al: Relationship between objective and subjective assessment of limb function in normal dogs with an experimentally induced lameness, *Vet Surg* 37:241-246, 2008.
8. Burton NJ, Owen MR, Colborne GR et al: Can owners and clinicians assess outcome in dogs with fragmented medial coronoid process? *Vet Comp Orthop Traumatol* 22:183-189, 2009.
9. Hielm-Bjorkman AK, Kapatkin AS, Rita HJ: Reliability and validity of a visual analogue scale used by owners to measure

chronic pain attributable to osteoarthritis in their dogs, *Am J Vet Res* 72:601-607, 2011.
10. Hudson JT, Slater MR, Taylor L et al: Assessing repeatability and validity of a visual analogue scale questionnaire for use in assessing pain and lameness in dogs, *Am J Vet Res* 65:1634-1643, 2004.
11. Hielm-Bjorkman AK, Kuusela E, Liman A et al: Evaluation of methods for assessment of pain associated with chronic osteoarthritis in dogs, *J Am Vet Med Assoc* 222:1552-1558, 2003.
12. Brown DS, Boston RC, Coyne JC et al: Development and psychometric testing of an instrument designed to measure chronic pain in dogs with osteoarthritis, *Am J Vet Res* 68:631-637, 2007.
13. Brown DS, Boston RC, Coyne JC et al: Ability of the Brief Canine Pain Inventory to detect response to treatment in dogs with osteoarthritis, *J Am Vet Med Assoc* 233:1278-1283, 2008.
14. Brown DS, Boston RC, Farrar JT: Comparison of force plate gait analysis and owner assessment of pain using the Canine Brief Pain Inventory in dogs with osteoarthritis, *J Vet Intern Med* 27:22-30, 2013.
15. DeCamp CE et al: Kinematic evaluation of gait in dogs with cranial cruciate ligament rupture, *Am J Vet Res* 57:120-126, 1996.
16. Millis DL, Hicks DA, Headrick JH et al: *Kinematic assessment of selected therapeutic exercises in dogs*. Proceedings of the 16th Annual American College of Veterinary Surgeons Symposium, Washington, DC, 2002.
17. Gassel AG, Millis DL, Schwartz P et al: *Kinematic gait analysis of the pelvic limb: comparison of overground versus incline walking in 10 dogs*. Proceedings of the 32nd Veterinary Orthopedic Society, Snowmass, CO, March 2005.
18. Headrick JH, Hicks DA, McEachern GL et al: *Kinematics of walking over cavaletti rails compared to overground walking in dogs*. Proceedings of the 2nd World Veterinary Orthopedic Congress/Veterinary Orthopedic Society, Keystone, CO, Feb 2006.
19. Hicks DA, Millis DL, Headrick JF et al: *Computer-assisted kinematic evaluation of compensatory movements of the lumbar spine associated with hind limb lameness in dogs with stifle osteoarthritis*. Proceedings of the 2nd World Veterinary Orthopedic Congress/Veterinary Orthopedic Society, Keystone CO, Feb 2006.
20. Stoller NB, Headick JH, Millis DL: *Kinetics of jumping dogs*. Proceedings of the 32nd Veterinary Orthopedic Society, Sun Valley, ID, March 2007.
21. Duant AM, Millis DL, Headrick JF: *Kinematics of stair ascent versus trotting in healthy dogs*. Proceedings of the 33rd Veterinary Orthopedic Society, Big Sky, MT, March 2008.
22. Headrick JF: *A description of the movement of the canine pelvic limb in three dimensions using an inverse dynamics method, and a comparison of two techniques to surgically repair a cranial cruciate ligament deficient stifle*. PhD Dissertation, University of Tennessee, 2012. http://trace.tennessee.edu/utk_graddiss/1471/
23. Ragetly CA, Griffon DJ, Mostafa AA et al: Inverse dynamics analysis of the pelvic limbs in Labrador retrievers with and without cranial cruciate ligament disease, *Vet Surg* 39:513-522, 2010.
24. Nielsen C, Stover SM, Schulz KS et al: Two-dimensional link-segment model of the forelimb of dogs at a walk, *Am J Vet Res* 64:609-617, 2003.
25. Chan CB, Spierenburg M, Ihle SL et al: Use of pedometers to measure physical activity in dogs, *J Am Vet Med Assoc* 226:2010-2015, 2005.
26. Hansen BD, Lascelles BDX, Keene BW et al: Evaluation of an accelerometer for at-home monitoring of spontaneous activity in dogs, *Am J Vet Res* 68:468-475, 2007.
27. Yam PS, Penpraze V, Young D et al: Validity, practical utility and reliability of Actigraph accelerometry for the measurement of habitual physical activity in dogs, *J Small Anim Prac* 52:86-92, 2011.
28. Brown DC, Boston RC, Farrar JT: Use of an activity monitor to detect response to treatment in dogs with osteoarthritis, *J Am Vet Med Assoc* 237:66-70, 2010.
29. How to track a lost animal. We tested three GPS devices for pets. *Consumer Reports,* May, 2012.
30. Jaegger G, Marcellin-Little D, Levine D: Reliability of goniometry in Labrador retrievers, *Am J Vet Res* 63:979-986, 2002.
31. Jaeger GH, Marcellin-Little DJ, DePuy V et al: Validity of goniometric joint measurements in cats, *Am J Vet Res* 68:822-826, 2007.
32. Peacock J, Millis DL, Weigel JP: *A comparative analysis of two methods of cranial drawer quantitation in the canine stifle*. Proceedings of the 26th Annual Conference of the Veterinary Orthopedic Society, Sun Valley, ID, 1999.
33. Edixhoven P et al: Accuracy and reproducibility of instrumented knee-drawer tests, *J Orthop Res* 5:378-387, 1987.
34. Doupe MB et al: A new formula for population-based estimation of whole body muscle mass in males, *Can J Appl Physiol* 22:598-608, 1997.
35. Daniel DM et al: Instrumented measurement of anterior laxity of the knee, *J Bone Joint Surg Am* 67:720-726, 1985.
36. Hewett TE, Noyes FR, Lee MD: Diagnosis of complete and partial posterior cruciate ligament ruptures: stress radiography compared with KT-1000 arthrometer and posterior drawer testing, *Am J Sports Med* 25:648-655, 1997.
37. Millis DL, Scroggs L, Levine D: *Variables affecting thigh circumference measurements in dogs*. Proceedings of the 1st International Symposium on Rehabilitative Physical Therapy Veterinary Medicine, 1999, p 157.
38. Martin AD et al: Anthropometric estimation of muscle mass in men, *Med Sci Sports Exerc* 22:729-733, 1990.
39. Ross M, Worrell TW: Thigh and calf girth following knee injury and surgery, *J Orthop Sports Phys Ther* 27:9-15, 1998.
40. Soderberg GL, Ballantyne BT, Kestel LL: Reliability of lower extremity girth measurements after anterior cruciate ligament reconstruction, *Physiother Res Int* 1:7-16, 1996.
41. Maylia E et al: Can thigh girth be measured accurately: a preliminary investigation, *J Sport Rehab* 8:43-49, 1999.
42. Doxey GE: Assessing quadriceps femoris muscle bulk with girth measurements in subjects with patellofemoral pain, *J Orthop Sports Phys Ther* 9:177-183, 1987.
43. Doxey GE: The association of anthropometric measurements of thigh size and B-mode ultrasound scanning of muscle thickness, *J Orthop Sports Phys Ther* 8:462-468, 1987.

44. Housh DJ et al: Anthropometric estimation of thigh muscle cross-sectional area, *Med Sci Sports Exerc* 27:784-791, 1995.
45. Knapik JJ, Staab JS, Harman EA: Validity of an anthropometric estimate of thigh muscle cross-sectional area, *Med Sci Sports Exerc* 28:1523-1530, 1996.
46. Howe TE, Oldham JA: The reliability of measuring quadriceps cross-sectional area with compound B ultrasound scanning, *Physiother Res Int* 1:112-126, 1996.
47. Weiss LW, Coney HD, Clark FC: Gross measures of exercise-induced muscular hypertrophy, *J Orthop Sports Phys Ther* 30:143-148, 2000.
48. Mitsiopoulos N et al: Cadaver validation of skeletal muscle measurement by magnetic resonance imaging and computerized tomography, *J Appl Physiol* 85:115-122, 1998.
49. Beneke R, Neuerburg J, Bohndorf K: Muscle cross-section measurement by magnetic resonance imaging, *Eur J Appl Physiol* 63:424-429, 1991.
50. Orima H, Fujita M: Computed tomographic findings of experimentally induced neurogenic muscular atrophy in dogs, *J Vet Med Sci* 59:729-731, 1997.
51. Walton JM, Roberts N, Whitehouse GH: Measurement of the quadriceps femoris muscle using magnetic resonance and ultrasound imaging, *Br J Sports Med* 31:59-64, 1997.
52. Visser M et al: Validity of fan-beam dual-energy X-ray absorptiometry for measuring fat-free mass and leg muscle mass: Health, Aging, and Body Composition Study—Dual-Energy X-ray Absorptiometry and Body Composition Working Group, *J Appl Physiol* 87:1513-1520, 1999.
53. Fuller NJ et al: Assessment of limb muscle and adipose tissue by dual-energy x-ray absorptiometry using magnetic resonance imaging for comparison, *Int J Obes Relat Metab Disord* 23:1295-1302, 1999.
54. Lieber RL et al: Growth hormone secretagogue increases muscle strength during remobilization after canine hindlimb immobilization, *J Orthop Res* 15:519-527, 1997.
55. Mayhew JL, Piper FC, Ware JS: Anthropometric correlates with strength performance among resistance trained athletes, *J Sports Med Phys Fitness* 33:159-165, 1993.
56. Gross MT et al: Relationship between multiple predictor variables and normal knee torque production, *Phys Ther* 69:54-62, 1989.
57. Hakkinen K, Komi PV, Kauhanen H: Electromyographic and force production characteristics of leg extensor muscles of elite weight lifters during isometric, concentric, and various stretch-shortening cycle exercises, *Int J Sports Med* 7:144-151, 1986.
58. Kanehisa H, Ikegawa S, Fukunaga T: Comparison of muscle cross-sectional area and strength between untrained women and men, *Eur J Appl Physiol* 68:148-154, 1994.
59. Arangio GA et al: Thigh muscle size and strength after anterior cruciate ligament reconstruction and rehabilitation, *J Orthop Sports Phys Ther* 26:238-243, 1997.
60. Maylia E et al: Can muscle power be estimated from thigh bulk measurements: a preliminary study, *J Sport Rehab* 8:50-59, 1999.
61. Young A et al: The effect of high-resistance training on the strength and cross-sectional area of the human quadriceps, *Eur J Clin Invest* 13:411-417, 1983.
62. Cooper H et al: Use and misuse of the tape-measure as a means of assessing muscle strength and power, *Rheumatol Rehabil* 20:211-218, 1981.
63. McCully K et al: The use of nuclear magnetic resonance to evaluate muscle injury, *Med Sci Sports Exerc* 24:537-542, 1992.
64. Burkholder WJ: Precision and practicality of methods assessing body composition of dogs and cats, *Compend Contin Educ Pract Vet* 23:1-10, 2001.
65. Brown DC, Boston RC, Coyne JC et al: A novel approach to the use of animals in studies of pain: validation of the canine brief pain inventory in canine bone cancer, *Pain Med* 10:133-142, 2009.
66. Hielm-Björkman AK, Rita H, Tulamo RM: Psychometric testing of the Helsinki pain index by completion of a questionnaire in Finnish by owners of dogs with chronic signs of pain by osteoarthritis, *Am J Vet Res* 70:727-734, 2009.
67. Conzemius, Michael G, Evans, Richard B: Caregiver placebo effect for dogs with lameness from osteoarthritis, *J Amer Vet Med Assoc* 241:1314-1319, 2012.
68. Reid J, Nolan AM, Hughes JML et al: Development of the short-form Glasgow Composite Measure Pain Scale.(CMPS-SF) and derivation of an analgesic intervention score, *Animal Welfare* 16(S):97-104, 2007.
69. Firth A, Haldane S: Development of a scale to evaluate pain in dogs, *J Amer Vet Med Assoc* 214:651-659, 1999.
70. Johnson DS, Smith RB: Outcome measurement in the ACL deficient knee—what's the score? *Knee* 8:51-57, 2001.
71. Wright RW: Knee injury outcome measures, *J Am Acad Orthop Surg* 17:31-39, 2009.
72. Noyes FR, McGinniss GH, Mooar LA: Functional disability in the anterior cruciate insufficient knee syndrome: review of knee rating systems and projected risk factors in determining treatment, *Sports Med* 1:278-302, 1984.
73. Noyes FR, Barber SD, Mangine RE: Abnormal lower limb symmetry determined by function hop tests after anterior cruciate ligament rupture, *Am J Sports Med* 19:513-518, 1991.
74. Olby NJ, De Risio L, Muñana KR et al: Functional scoring system in dogs with acute spinal cord injuries, *Am J Vet Res* 62:1624-1628, 2001.
75. Levine GJ, Levine JM, Budke CM et al: Description and repeatability of a newly developed spinal cord injury scale for dogs, *Prev Vet Med* 89:121-127, 2009.
76. Budke CM, Levine JM, Kerwin SC et al: Evaluation of a questionnaire for obtaining owner-perceived, weighted quality-of-life assessments for dogs with spinal cord injuries, *J Am Vet Med Assoc* 233:925-930, 2008.
77. Levine JM, Budke CM, Levine GJ et al: Owner-perceived. Weighted quality of life assessments in dogs with spinal cord injuries, *J Am Vet Med Assoc* 233:931-935, 2008.
78. Freeman LM, Rush JE, Farabaugh AE et al: Development and evaluation of a questionnaire for assessing health-related quality of life in dogs with cardiac disease, *J Am Vet Med Assoc* 226:1864-1868, 2005.
79. Wiseman-Orr ML, Nolan AM, Reid J et al: Development of a questionnaire to measure the effects of chronic pain on health-related quality of life in dogs, *Am J Vet Res* 65:1077-1084, 2004.

14 Rehabilitating the Painful Patient: Pain Management in Physical Rehabilitation

Steven M. Fox and Robin Downing

Divinum est opus sedare dolorum.
(Divine is the work to subdue pain.)

Galen

Although pain in canine patients is inevitable in the face of surgery, trauma, injury, osteoarthritis, and so on, suffering is optional. It is well established that uncontrolled pain is not only ethically problematic, but physiologically damaging as well.[1] In recent years pain management modalities have been developed in human medicine to address the evolving understanding of the complexities of the pain experience. Many of these pain treatment strategies have been imported into veterinary medicine. Pain and suffering should be sought out, the cause identified to the greatest degree possible, and alleviated when feasible.

Canine physiotherapy/rehabilitation is a discipline that encompasses the application of physical therapy techniques (developed in humans) to dogs whose comfort and function have been compromised in some way. A wide variety of modalities and methods can be applied, from the most highly technical and advanced techniques used for complex orthopedic surgery recoveries, to simple therapeutic techniques that can be taught to pet owners for use at home. The overall goal is to restore, maintain, and promote optimal function, optimal fitness, wellness, and quality of life as they relate to movement disorders and health.

Often the patient who presents for rehabilitation is reluctant to move and exercise. This reluctance may be caused by the patient's unwillingness, inability, or a combination of these factors. It is important to understand that most of the dogs who are candidates for physiotherapy and rehabilitation are painful. It is equally important to remember that as veterinary health care providers, we have an obligation to provide compassionate care. If the pain of a rehabilitation patient is not appropriately or adequately addressed, then physiotherapy will be problematic. Consequences of unresolved pain may include:

- Unnecessary suffering for the painful patient
- Tissues that are unable to heal normally
- Restriction of range of motion and other tissue dysfunction from the influence of pain signals
- Undermining the trust inherent in the family–pet relationship
- Tissue contraction/atrophy from disuse driven by pain
- Resentment on the part of the patient of all caregivers
- The transition/transformation from adaptive pain that serves a purpose to a maladaptive pain state in which the pain itself is pathologic

An underlying tenet of veterinary medicine is "Do No Harm," and this applies equally to physical rehabilitation. Fundamental to the success of any physical rehabilitation regimen is the control of pain. And fundamental to the success of controlling pain is the understanding that a multifaceted (also called multimodal) approach appears to provide optimal outcome. The goal for successful integration of pain management and physical rehabilitation is to build a pain management plan that allows for maximum comfort to open the door for maximum movement.

Pain Processes

There are excellent sources for in-depth discussions of pain physiology and pathophysiology.[2-4] Within the context of physical rehabilitation it is important to have an understanding of basic pain processes, as this information will influence treatment choices for individual patients. For instance, patients experiencing acute pain following orthopedic surgery have different needs than the elderly dog experiencing the chronic maladaptive pain associated with long-standing osteoarthritis. Practitioners must appreciate the differences among patients and their pain experiences so that they may create individualized pain management strategies. Likewise, practitioners must understand properties unique to different drugs and drug classes as well as these drugs' mode of action to create a targeted approach to pain.[5]

Adaptive and Maladaptive Pain

Pain can be divided into two general categories recognized as adaptive and maladaptive. Adaptive pain protects the body from injury and promotes healing by inhibiting activity when injury has occurred. In contrast, maladaptive pain reflects pathologic activity of the nervous system—the

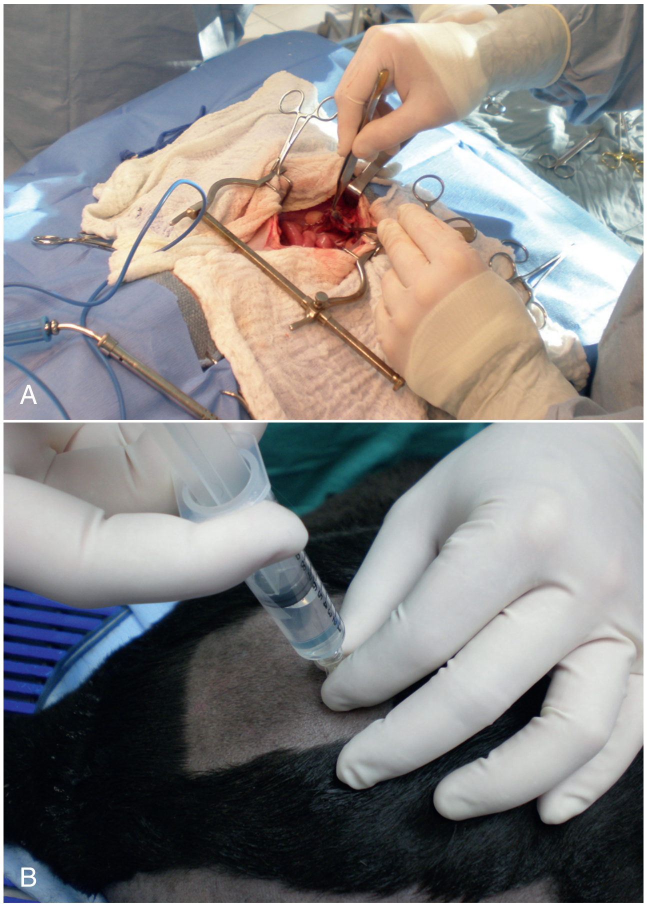


Figure 14-1 Examples of acute pain: surgery (*A*) or a needle insertion (*B*).

result of complex processes of neuronal plasticity—pain as a disease state.[5,6]

Adaptive, everyday acute pain (Figure 14-1) is also known as nociceptive or physiologic pain and occurs when noxious stimuli (mechanical, thermal, or chemical) are transduced into electrical signals that are then transmitted to the spinal cord. These signals undergo modulation in the spinal cord before being relayed to the brain, where the conscious brain interprets these transmissions, resulting in pain perception. Adaptive pain is purposeful. Dr. Frank Vertosick states, "Pain is a teacher, the headmaster of nature's survival school."[7] Acute pain tends to be of relatively short duration and, if addressed properly, is relatively easy to treat.

Chronic maladaptive pain may result from sustained noxious stimuli such as ongoing inflammation or profound tissue damage, or it may be present independent of the inciting cause. Regardless of its etiology, maladaptive pain offers no or little useful biologic function or survival advantage. The nervous system itself becomes the source of the pathology and contributes to patient morbidity. Effective treatment for maladaptive pain can be an enigma. A number of studies have shown that the longer a pain lingers, the harder it is to eradicate because of resultant changes to both the peripheral and central nervous system.

Chronic pain was traditionally defined as pain lasting more than 3 or 6 months, depending on the source of the definition.[8,9] More recently, chronic pain has been defined as "pain that extends beyond the period of tissue healing and/or with low levels of identified pathology that are insufficient to explain the presence and/or extent of pain."[10] There is no general consensus on the definition of chronic pain. In clinical practice it is often difficult to determine when acute pain has become chronic—when adaptive pain crosses the threshold to become maladaptive.

Adaptive to Maladaptive Pain

A progression from adaptive to maladaptive pain can be described as a spectrum composed of three major stages or phases of pain, suggesting that different neurophysiologic mechanisms are involved, depending on the nature and time course of the originating stimulus.[11] These three phases are (1) the processing of a brief noxious stimulus, (2) the consequences of prolonged noxious stimulation, leading to tissue damage and peripheral inflammation, and (3) the consequences of neurologic damage or changes, including peripheral neuropathies and central pain states. This spectrum of architectural change within the nervous system and the resultant pain pathology may be represented graphically as in Figure 14-2.

The journey of the nervous system toward maladaptive pain is often described as windup. Windup describes activity-dependent plasticity characterized by a progressive increase in action potential output from dorsal horn neurons elicited during the course of prolonged, repeated low-frequency C-fiber or nociceptor stimuli.[12,13] One receptor critical to this process is the *N*-methyl-D-aspartate (NMDA) receptor. NMDA receptors are only activated in the face of prolonged depolarization such as occurs with chronic pain.[14] Their activation can cause neural cells to sprout new connective endings,[15] thus expanding the receptive fields of the painful patient. Accompanying windup, less glutamate is required to transmit the pain signal and more antinociceptive input is required for analgesia. The complexity of maladaptive pain and the phenomenon of windup continues to provoke ongoing study. Studies have implicated activated microglia, norepinephrine (NE) levels, serotonin (5-HT) activity, as well as transient receptor potential (TRP) receptors as playing a role in maladaptive pain.

The rehabilitation practitioner must grasp the concept of this transition from adaptive to maladaptive pain in order to understand how best to address the pain management needs of the patient that is presented for physiotherapy. There are multiple potential targets of analgesic treatment in the patient experiencing both adaptive and maladaptive pain. It is appropriate, when putting together the pain management strategy for a specific patient, to address whatever targets

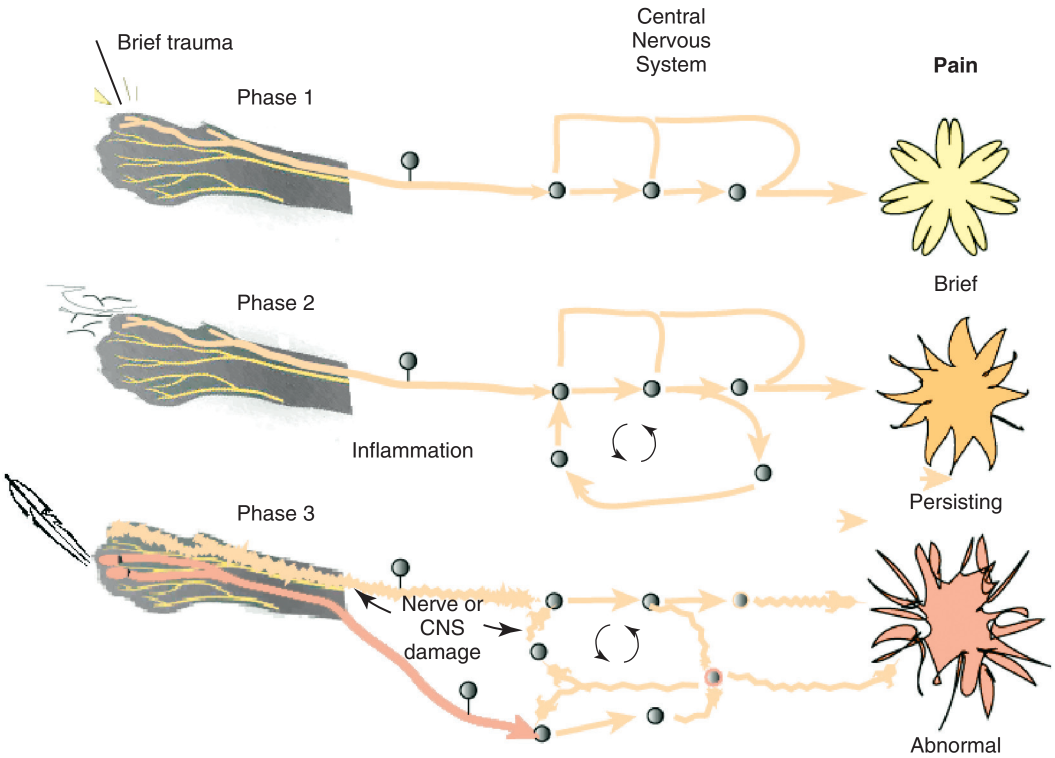


Figure 14-2 Proposed pain states as pain transitions from adaptive to maladaptive.

of therapy may be relevant. The rehabilitation practitioner *must* work in tandem with the primary care practitioner, the surgeon (when appropriate), and the dog's human companions to facilitate open and reciprocal communication about the patient's pain status at every stage of treatment. Some patients require a very complex and aggressive pain management protocol to accomplish the goals of physical rehabilitation. As physiotherapy progresses, and the patient improves, the pain management strategy must be adjusted to meet those changing needs. Likewise, if the patient worsens during rehabilitation, reassessment of the patient and revision of its pain protocol are essential. Without some understanding of the complexities of long-standing pain, and of treating pain as a disease in and of itself, the rehabilitation patient may suffer needlessly.

Putting Together a Pain Management Plan for Surgery Patients

Postsurgical Pain and Rehabilitation

Whenever possible, the rehabilitation practitioner should be in contact with the primary care practitioner and surgeon before a surgical patient is referred for physiotherapy. This allows for dialog about the patient's diagnosis, the surgical plan, and the pre-, intra-, and postoperative pain management strategy. It also allows the rehabilitation practitioner to design the postoperative physiotherapy program and implement any appropriate techniques immediately after surgery—either at the surgical practice or in the home during the initial postoperative healing period. It also allows the rehabilitation practitioner the potential to play a role in preventing the patient from evolving into a maladaptive pain state.

Preemptive Analgesia

Pain memories imprinted within the CNS, mediated by NMDA receptors, produce hyperalgesia and contribute to allodynia—pain that is produced by a stimulus that does not normally provoke pain. A preoperative blockade of nociception may prevent postsurgical wound pain or pain hypersensitivity following surgery. This is the concept of preemptive analgesia articulated by the eminent pain physiologist Patrick Wall.[16] In 1988, Professor Patrick Wall introduced the concept of preemptive analgesia to clinicians with his editorial in the journal *Pain*.[17] Successful preemptive analgesia must meet three criteria. It must be (1) intense enough to block all nociception, (2) wide enough to cover the entire surgical area, and (3) prolonged enough to last throughout surgery and even into the postoperative period.

General anesthesia with an inhalant agent (e.g., isoflurane, sevoflurane) does *not* prevent central sensitization. The potential for central sensitization exists even in unconscious patients who are unresponsive to surgical stimuli.[18,19] Noxious stimuli still enter the spinal cord during apparently adequate anesthesia.[20] The purpose of preemptive analgesia is to prevent sensitization of the nervous system throughout the perioperative period. Pain is to be expected from an initial surgery and the hypersensitivity that subsequently develops. Analgesia administered after sensitization may decrease pain somewhat, but has little long-term benefit in addressing the pain resulting from postsurgical inflammation. Analgesia administered before surgery limits inflammatory pain and decreases subsequent hypersensitivity. The most effective preemptive analgesic regimen occurs when it is initiated before surgery and continued throughout the postoperative period. Equally important is to use a complement of medications that work synergistically; for example, an opioid, alpha-2 agonist, NMDA antagonist, local anesthesia, and an NSAID.[21-23] It bears reminding that good nursing care in the immediate postoperative period can contribute to the comfort of the surgical patient.[24]

Postoperative Analgesia

For the majority of rehabilitation patients that have experienced surgery, NSAIDs form the foundation of pain management in the postoperative period. Antiinflammatory drugs play such a substantial role in perioperative pain management[25] because surgery cannot be performed without subsequent inflammation. Reducing the inflammatory response in the periphery, and thereby decreasing sensitization of the peripheral nociceptors, should attenuate central sensitization.[26] In human medicine, the use of perioperative NSAIDs has reduced the use of patient-controlled analgesic morphine by 40% to 60%.[27,28]

Depending upon the patient's needs, an NSAID may be adequate as a stand-alone pain management agent. However, the rehabilitation practitioner must be especially vigilant about pain issues in the immediate postoperative period. Inadequately managed postoperative (adaptive) pain can transition to maladaptive pain, creating additional problems for the physiotherapy patient. A postoperative rehabilitation patient should be assessed for pain regularly, both at the time of therapy and through conversation and questioning with the pet owner to evaluate activities of daily living. One source of methods for scoring and evaluating pain may be found in the AAHA/AAFP Pain Management Guidelines for Dogs and Cats.[29] It may be that a postsurgical patient may need one or more of the adjunctive pain management strategies outlined in the following chronic pain section.

The Elements of a Comprehensive Pain Management Plan for Chronic Pain Patients

Pharmaceutical Pain Management for the Maladaptive Pain Patient

A majority of nonsurgical patients presented for physical rehabilitation are dogs suffering from chronic, maladaptive

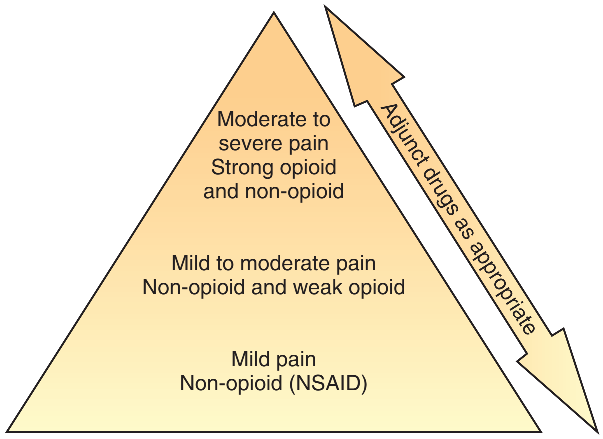


Figure 14-3 The World Health Organization Cancer Pain Ladder. (Adapted from WHO 31.)

pain—often as a result of the degenerative changes that characterize osteoarthritis. Regardless of the underlying etiology, maladaptive pain patients present unique challenges to the rehabilitation practitioner, the primary care provider, and the pet owner who has witnessed the erosion of the dog's activities of daily living. The goal is to break the cycle of maladaptive pain to allow the patient to engage in physical rehabilitation activities designed to restore function, build strength, and ultimately minimize the use of pharmacologic agents to sustain comfort.

When building a comprehensive pharmacologic pain management plan for these patients, it is instrumental to consider the World Health Organization (WHO) Cancer Pain Ladder as a model (Figure 14-3). The principle articulated in this model is to begin pain management using medications that are effective against mild to moderate pain—generally NSAIDs—and building upon that foundation by adding adjunctive medications that attack maladaptive pain via synergistic pathways/modes of action as well as nonpharmacologic strategies. The result is a pyramid in which each addition builds upon the last layer of pain management, providing a foundation for the next layers to be added. The WHO ladder for treating cancer pain in humans[30] has been adopted for a variety of different types of pain in both humans and veterinary patients.

As veterinary pain management becomes more sophisticated, pharmacologic combination therapies from human medicine are being considered and implemented. It is much easier to identify maladaptive pain states in humans than in nonhuman species. Figure 14-4 suggests a logical approach to managing the progressive pain state, wherein pain management uses a multimodal scheme in which drug classes are added rather than substituted. The doses are empiric, and one could debate the order of implementation from bottom to top. The merit of this approach is the addition of agents from different drug classes, attempting to block as many of the pain pathways (transduction, transmission, modulation, and perception) as possible. The

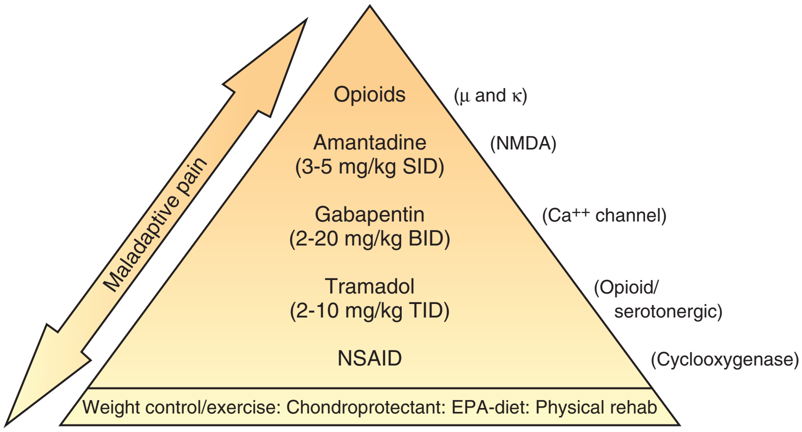


Figure 14-4 Proposed analgesic ladder for progressing pain states using drugs with different (complementary) modes of action. (Doses are empiric, and combination efficacy and safety data are unexplored.)

illustration also acknowledges the important role of nonpharmacologic contributions to pain management. Care must be taken as clinicians await clinical trials that clearly and specifically provide information about (1) each drug's mode of action, (2) the pharmacokinetics in the target species, and (3) the physiology as well as pathophysiology in the veterinary target species. At the moment, only a handful of studies may be cited in support of the use of several of the adjunctive agents that are enjoying increasing use in veterinary patients.[31-34]

Before beginning any multimodal pharmacologic pain management strategy for the rehabilitation patient with maladaptive pain, it is important to ensure due diligence to uncover any existing disease or co-morbidity that could influence decisions about medications. It is also imperative to understand the potential interactions among medications that may be chosen to manage maladaptive pain, as these drugs will presumably be given over long periods—potentially months to years.

Pretreatment screening should include a thorough physical examination, including a review of the patient's medical history and medical record for any previous or concurrently administered medications. A metabolic profile to evaluate renal and hepatic function and a hemogram are fundamental. A more sensitive way to assess for potential renal compromise is to include a test for microalbuminuria[35]—a marker for increased permeability of the basement membrane of the glomerulus. Although not pathognomonic for chronic renal failure, microalbuminuria alerts the practitioner to the need for additional vigilance of renal function. There is no consensus regarding the frequency of monitoring metabolic and organ function in the face of chronic-use medications for pain; however, a 2- to 4-month interval for monitoring is reasonable in most older dogs.

Nonsteroidal Antiinflammatory Drugs

Nonsteroidal antiinflammatory drugs remain the cornerstone for treating pain associated with degenerative joint disease (osteoarthritis; OA) and other sources of chronic pain. Nonsteroidal antiinflammatory drugs have effects at numerous locations along nociceptive pathways, both peripherally and centrally. This class of drugs manifests antiprostaglandin effects peripherally, but they are also considered central-acting because of their inhibition of cyclooxygenase within the dorsal horn of the spinal cord. Cyclooxygenase-2 (COX-2)–mediated PGE_2 production contributes to a late-onset, prolonged, and diffuse phase of central sensitization.[36] Cyclooxygenase-2-specific NSAIDs dampen central sensitization, and therefore play an important role in a multimodal pain management strategy. Cyclooxygenase-1–sparing NSAIDs (Deracoxib,[37] meloxicam, and firocoxib[38]) have their optimal theoretical advantage where there is concern for platelet aggregation, or homeostasis.

The integration of an NSAID in a pain management protocol is particularly appealing when considering the theory by some[39] that the origin of all pain is inflammation and the inflammatory response. However, it must be appreciated that the ceiling effect of NSAIDs can limit their analgesic potential, thereby suggesting they are best, when used alone, for mild and moderate pain. Among a population of canine pain patients, all NSAIDs are effective, but a practitioner may witness variable responses to NSAIDs among individuals.

Tramadol

Tramadol has become a popular analgesic adjunct over the past decade, having gained a worldwide reputation as an effective, safe, and well-tolerated drug for inhibition of moderate pain in humans. It has come to fill the gap on the WHO analgesic ladder between the weak analgesics (NSAIDs) and strong opioids (e.g., morphine). It is not classified as a true opioid, but does have a weak affinity for μ-receptors. In addition, tramadol inhibits both serotonin and norepinephrine reuptake.

Adverse effects include sedation, dysphoria, vomiting, diarrhea, and inappetence. Tramadol tablets are extremely bitter, so care must be taken to hide the tablet to prevent the dog from tasting it and becoming food averse. Because of its effects on serotonin reuptake, concomitant use of a selective serotonin reuptake inhibitor (SSRIs such as tricyclic antidepressants) or monoamine oxidase inhibitor (MAOI) can lead to serotonin syndrome. The signs of serotonin syndrome include: agitation, confusion, rapid heart rate, dilated pupils, shivering, high fever, seizures, and unconsciousness. Serotonin syndrome is potentially life-threatening.

Doses range from 2 to 10 mg/kg po tid. This is empiric dosing, as tramadol is not approved for dogs or cats. Only oral preparations are available in the United States. The pharmacokinetics of tramadol have been studied in dogs, and it appears that the half-life is quite short, approximately 1.7 hours, necessitating frequent dosing.[40] Evaluation of an oral sustained-release formulation of tramadol did not yield positive results in the dog.[34] Within the veterinary profession, there are not yet dose titration studies in our target species. To date there are no published dose titration studies or studies of safety or efficacy in dogs or cats.

Anticonvulsants

The anticonvulsant gabapentin is structurally analogous to γ-aminobutyric acid (GABA) and was initially used as an antiepileptic, particularly in children. Gabapentin is widely used in human medicine for its activity in certain maladaptive pain syndromes including postherpetic neuralgia, painful diabetic peripheral neuropathy, and fibromyalgia. Gabapentin is theorized to work by binding to the α_2-δ_1 subunit of presynaptic voltage-gated calcium channels that are upregulated during central sensitization. Its analgesic effect may well be proportional to the magnitude of sensitization within the dorsal horn.[41,42]

Gabapentin may be most useful as an adjunct to other analgesics, particularly in those patients suspected of experiencing maladaptive pain. One study in rats suggested that gabapentin may be analgesic in the pain associated with osteoarthritis.[43] Gabapentin use in human pain medicine has recently extended into the management of more acute conditions, including an adjunctive role in the perioperative period for the prevention of persistent postoperative pain.[44] The liquid form of gabapentin, labeled for human use, should not be used in pets, as it contains xylitol, which is hepatotoxic in dogs. Gabapentin can be compounded into liquid form for veterinary use, however.

Simultaneous administration of oral antacids may decrease the bioavailability of gabapentin. They should be given at least 2 hours apart. Also, coadministration of morphine may increase gabapentin's effects and increase the likelihood of side effects such as sedation. The use of gabapentin in three dogs suspected of experiencing neuropathic pain, a maladaptive pain state, has been documented.[33] Additional studies need to be conducted in veterinary species.

In dogs clinicians are using gabapentin as an adjunct to other therapies for those patients suspected of experiencing maladaptive pain.[33] Reported starting doses are 5 to 10 mg/kg po bid-tid. The most common adverse effect of gabapentin in humans is dose-dependent sedation, and in clinical practice this appears to be the case for dogs. For most patients a downward dose adjustment resolves the issue. In dogs with maladaptive pain, positive results to the addition of gabapentin may manifest in as little as 48 to 72 hours. As stated the analgesic effect of gabapentin may be proportional to the level of maladaptive pain the patient is experiencing. Thus the dose of gabapentin may be increased to meet the need of the patient, with sedation as the primary dose-limiting effect. Gabapentin should not be discontinued abruptly because withdrawal may precipitate rebound pain. Once the dog's pain is well managed the dosage can be slowly reduced over a period of weeks to months.

N-*Methyl-D-Aspartate Antagonists*

The NMDA receptor plays a major role in central sensitization[45]; therefore inhibition of this receptor validates the implementation of an NMDA-antagonist in an analgesic protocol. Amantadine is an NMDA receptor antagonist and acts through stabilization of NMDA channels in the closed state.[46] Amantadine has been used to treat neuropathic pain in humans. A clinical trial was reported in which dogs with refractory osteoarthritis received amantadine as an adjunctive therapy in addition to an NSAID.[32] The conclusion was that amantadine performed well in improving physical activity in the treated dogs. Toxicity studies have been performed,[31] and a dose of 8 mg/kg once daily was not implicated in adverse signs.

As an adjunct to an NSAID in the dog with chronic OA pain, a dose of 3 to 5 mg/kg per day po may be given. In the author's (Downing) experience, this drug and dose is extremely well tolerated. Amantadine is available commercially as 100 mg capsules and as a 10 mg/mL oral syrup.

Tricyclic Antidepressants

Tricyclic antidepressants (TCAs) are characterized by their multiple modes of action, with a particular ability to inhibit reuptake of monoamines (serotonin and norepinephrine) from presynaptic terminals. Additionally TCAs block several receptors (cholinergic, adrenergic, histaminergic) and ion channels, including Na^+ channels. The mechanism of TCA action might best be identified as five drugs in one: serotonin reuptake inhibitors, norepinephrine reuptake inhibitors, anticholinergic-antimuscarinic drugs, α-1 adrenergic antagonists, and antihistamines. These drugs have been used in humans to treat maladaptive pain at much lower doses than those used to treat depression.

Longer-term administration alters the receptor binding of NMDA. Tricyclic NMDA activity is likely analgesic via its inhibition of neuronal hyperexcitability.[47] Alpha-adrenergic-receptor blockade in peripheral neuropathy may be relieving pain generated or maintained by noradrenergic stimulation of highly sensitive receptors such as those identified on sprouts from diseased peripheral nerves.[48] Tricyclic antidepressants may stabilize both diseased peripheral nerves and hyperexcitable neurons of the central nervous system (CNS) by blocking sodium channels.[49]

Although TCAs may be effective as an adjunctive therapy for maladaptive pain in animals, they have not yet been evaluated. The TCAs should not be used concomitantly with drugs that affect serotonin levels such as tramadol.

Orally Administered Opioids

Although controversial, studies suggest that maladaptive pain secondary to peripheral nerve damage, more often than not, shows reduced sensitivity to opioids.[50] This is attributable to a reduction in spinal opioid receptors, non-opioid receptor-expressing Aβ fiber-mediated allodynia, increased cholecystokinin antagonism of opioid actions, and NMDA-mediated dorsal horn neuronal hyperexcitability, likely requiring a greater opioid inhibitory countereffect. Another problem with oral opioids in dogs is the very low (methadone <30%; morphine 5%) bioavailability[51,52] because opioids are metabolized in the liver. Codeine has the highest bioavailability, approaching 60%,[53] and is often administered in combination with acetaminophen (***acetaminophen is lethal in cats***). Although morphine and hydromorphone are available as suppositories, there appears to be little difference in efficacy or bioavailability (≈20%) from oral administration.[54] However, this delivery form is an option if oral administration is unavailable. Although injectable opioids remain the gold standard for managing severe acute pain, oral opioids seem lacking in their application (orally) in most animals with chronic maladaptive pain.

Nonpharmacologic Approaches to the Chronic Pain Patient

Acupuncture

Acupuncture falls under the category of complementary and alternative medicine (CAM) as part of traditional Chinese medicine (TCM) and is used in humans in at least 78 countries worldwide.[55] There are sound physiologic mechanisms for the analgesic effects of this treatment.[56] A common feature shared by all styles of acupuncture is the use of needles to initiate changes in the soft tissue. Needles and needle-induced changes are believed to activate the built-in survival mechanisms that normalize homeostasis and promote self-healing. In this context acupuncture can be defined as a physiologic therapy coordinated by the brain that responds to the stimulation of manual or electrical needling of peripheral sensory nerves, in which acupuncture does not treat any particular pathologic system, but normalizes physiologic homeostasis and promotes self-healing.[57] Acupuncture can be effective for both peripheral soft tissue pain and internal disorders, but in the case of peripheral soft tissue pain the result appears more predictable because of the local needle reaction.

Hypothesis of Acupuncture Analgesia Mechanisms

The leading hypotheses of acupuncture analgesia mechanisms include local effects of needling, neuronal gating, the release of endogenous opiates, and the placebo effect. It is further proposed that the CNS is essential for the processing of these effects via its modulation of the autonomic nervous system, the neuroimmune system, and hormonal regulation. Clinical observation suggests that acupuncture needling may achieve at least four therapeutic goals:

1. Release of physical and emotional stress
2. Activation and control of immune and antiinflammatory mechanisms
3. Acceleration of tissue healing
4. Pain relief secondary to endorphin and serotonin release.

Acupuncture therapy is considered to activate built-in survival mechanisms (e.g., self-healing potential). Therefore it is effective for those symptoms that can be completely or partially healed by the body. Additionally each individual has a different self-healing capacity influenced by genetic makeup, medical history, lifestyle, and age, all of which may be dynamically changing.[58] Acupuncture can be used to release trigger or tender points in addition to its application for musculoskeletal pain.

Acupuncture can be initiated immediately for the benefit of the rehabilitation patient experiencing maladaptive pain. In this context, acupuncture may be used to augment the effects of the pharmacologic intervention that is chosen. It is the author's experience (Downing) that acupuncture often results in reduced doses of medications needed to break the cycle of a rehabilitation patient's maladaptive pain.

Other Therapies

Other techniques exist for augmenting the pharmacologic intervention on behalf of a painful dog presented for rehabilitation. These include the application of low-level laser therapy (LLLT), also referred to as cold laser, therapeutic laser, extracorporeal shock wave therapy, therapeutic ultrasound, cryotherapy, heat therapy, and electrical stimulation. These modalities are described in detail elsewhere in this book and by others.[59-61]

Environmental Modification

It is difficult to overstate the positive difference that simple environmental changes can make for the maladaptive pain patient. Yet, this is one area that is easy to overlook because it does not involve as much hard science as choosing a pharmacologic regimen or calculating an appropriate nutritional profile. With a few simple questions it is fairly straightforward to determine if the rehabilitation patient will benefit from environmental modification. It does mean taking the initiative to gather information about the home, such as floor surfaces, stairs, placement of food and water dishes, type of bedding, location of bedding, and so on.

One of the simplest environmental modifications that can have a positive effect on the chronically painful rehabilitation patient is to protect the dog from slippery floor surfaces. Aging patients, particularly those with OA, experience a loss of proprioceptive function in joint receptors versus younger patients.[62,63] Nonskid area rugs, flooring used in children's play areas, and rubber-backed mats are some examples of ways a dog owner can make the home more comfortable for the pain patient.

Raising food and water dishes to between elbow and shoulder height makes for more comfortable meal times. Be sure the dog is able to stand on a nonskid surface while eating and drinking. Have the client explore the home for potential problem spots. Steps in and out of the house, patio stones, and garage floors can create unintentional challenges for the painful dog. Ramps for getting into and out of vehicles should be recommended. Likewise, have the client consider ramps or steps if the dog is used to getting onto and off furniture. It may be best to use child restraint gates at the top and bottom of staircases to prevent unsupervised access. Memory foam or eggshell foam may be a more comfortable sleeping surface for the painful dog. Assistive devices like slings and wheelchairs can sustain mobility during the initiation of appropriate multimodal pain management strategies. Finally it is best for the dog to receive moderate exercise every day than to do excessive exercise on the weekend, requiring the rest of the week to recover.

Putting It All Together for the Painful Patient

In human pain medicine, pain practitioners understand that pain is what the patient says it is, and that their obligation is to believe the patient. In veterinary medicine, the practitioner, whether the primary care provider, the surgeon on an orthopedic case, or the rehabilitation practitioner, has an obligation to advocate on behalf of a patient that cannot advocate for itself, and that means actively looking for pain in the patients we see. Assessing nonhuman patients presents its own challenges; however, these challenges must not prevent caretakers from taking an aggressive approach to pain, particularly to the long-standing suffering of the maladaptive pain patient. For assistance in choosing a pain scoring system, the reader is referred to the AAHA/AAFP Pain Management Guidelines for Dogs and Cats,[29] and the International Veterinary Academy of Pain Management (www.ivapm.org).

There are many things to consider when confronted with a painful rehabilitation patient. The most important first step is recognizing that rehabilitation will be counterproductive if the patient's pain is not identified and managed appropriately. Pain is a subjective, individual phenomenon, and there is tremendous variability among individual patients. For these reasons, when in doubt, it is best to treat. Often after pain management is initiated the client will report a normalization of the dog's behaviors and activities of daily living. It is common to hear a client say, "I didn't know he was painful."

Considering that each patient is unique, an individual pain management protocol should be designed to accompany each patient's rehabilitation plan. The pharmacologic component of that protocol is based on assessment of the individual animal and the particular rehabilitation treatment/session. For immediate postoperative rehabilitation the importance of a comprehensive multimodal, perioperative analgesic protocol cannot be overstated because of its impact on the postoperative recovery period. For the dog with chronic maladaptive pain, monotherapy with a single-agent like an NSAID is likely to fail.

How does a rehabilitation practitioner begin putting together a multimodal pain management strategy? Earlier in this chapter the analogy was drawn between building a pain management plan and building a pyramid. This provides us with a visual expression of our task on behalf of the painful rehabilitation patient. Because each patient is an individual, each pain management protocol must be created with that patient's individual needs in mind. However, there are some general principles to consider based on the author's (Downing) experience in a pain management referral practice:

1. Start with a complete and thorough physical examination, including a neurologic evaluation, orthopedic evaluation, and careful palpation for pain.
2. Spend time talking with the client to understand the dog's activities of daily living, the daily household routine, and the home environment (stairs, floor surfaces, placement of food and water dishes, etc.).
3. Conduct a complete metabolic profile to assist in the choice of medications as well as to identify any co-morbidities. Treat any co-morbidities (e.g., hypothyroidism) concurrently.
4. Focus on BCS (body condition score). If the dog is overweight this *will* interfere with pain management and subsequent rehabilitation.[64,65] Create a nutritional plan to assist the dog in achieving a normal BCS.
5. For those dogs with OA provide an EPA-supplemented nutritional product.[66-68]
6. Create a pharmacologic intervention plan. If NSAIDs are appropriate for the patient, based on the metabolic profile, prescribe one. For a maladaptive pain patient, gabapentin is a reasonable addition in the first round of therapy.
7. If possible consider acupuncture, laser therapy, cryotherapy, or another modality to provide relief.

8. Create expectations for the client so he or she can appropriately gauge the success or failure of each step in the treatment progression.
9. Create a written plan for the client, the medical record, and the primary care provider to minimize misunderstanding as well as the risk for medical error. Painful rehabilitation patients require complex care and monitoring, so make it as easy for the client as possible.
10. Schedule each pain reassessment for chronic pain patients *before* the client leaves the facility. The first pain reassessment should be performed 7 to 10 days after initiating treatment. Subsequent reassessments should be scheduled at 10- to 14-day intervals until the dog's pain is well managed. Although we should not increase the dose of an NSAID above the recommended dosing, it is common to increase the dose of gabapentin to meet the patient's need. For patients with long-standing maladaptive pain, it is not uncommon to need gabapentin doses approaching 30 to 40 mg/kg bid-tid. Patients being treated for acute post-operative pain should be assessed several times per day for adequate comfort.
11. If the patient remains painful at the third visit following an increase in gabapentin dosing, consider adding amantadine, and schedule the next recheck in 7 to 14 days, depending on the patient's level of discomfort. The greater the discomfort, the shorter should be the recheck interval. If needed, tramadol may be used for episodes of breakthrough pain. Likewise, a TCA may be used for refractory pain. Do not use tramadol and a TCA together.
12. After the dog's pain is reasonably controlled rehabilitation can begin in earnest.
13. Schedule appropriate metabolic monitoring based on the patient and the medications chosen.
14. After pain is well managed and the dog is actively engaged in rehabilitation activities, coordinate the titration of medication doses to achieve minimum effective doses, recognizing that some individuals may need multiple medications long term.

The goal as practitioners is to achieve and maintain a delicate balance with painful patients to keep them as comfortable, active, mobile, and functional as possible to sustain the precious family–pet relationship. After a pain management strategy is under way remember the "three Rs"—Recheck, Reassess, and Revise. Painful rehabilitation patients provide a moving target for our care. Remain in dialog with the clients to notice patterns of behavior that might precipitate modification of the treatment plan. Daily journals of the dog's activities and interactions with the family can provide important insights.

Our obligation to our painful rehabilitation patients is to advocate on behalf of creatures who cannot advocate for themselves. We need to practice compassionate care medicine. Animals must have their pain relieved for their physiotherapy to be maximally effective.

REFERENCES

1. Rollin BE: *The ethics of pain management. Handbook of veterinary pain management*, ed 2, St. Louis, 2009, Mosby Elsevier.
2. Mathews KA: *Vet Clin North Am Sm Anim Pract* 30:4, 2000.
3. Mathews KA: *Vet Clin North Am Sm Anim Pract* 38:6, 2008.
4. Gaynor JS, Muir WW, editors: *Handbook of veterinary pain management*, ed 2, St. Louis, 2009, Mosby Elsevier.
5. Woolf CJ: Pain: moving from symptom control toward mechanism-specific pharmacologic treatment, *Ann Int Med* 140:441-451, 2004.
6. Mathews KA: Neuropathic pain in dogs and cats: if only they could tell us if they hurt, *Vet Clin North Am Sm Anim Pract* 38(6):1365-1414, 2008.
7. Vertosick FT: *Why we hurt. The natural history of pain*, New York, 2000, Harcourt.
8. Russo CM, Brose WG: Chronic pain, *Annu Rev Med* 49:123-133, 1998.
9. Schaible HG, Richter F: Pathophysiology of pain, *Langerbecks Arch Surg* 89:237-243, 2004.
10. Jacobsen L, Mariano A: General considerations of chronic pain. In Loeser JD, Butler SH, Chapman SR, editors: *Bonica's management of pain*, ed 2, Baltimore, 2001, Lippincott Williams & Wilkins.
11. Cervero F, Laird JM: From acute to chronic pain: peripheral and central mechanisms. In Bountra C, Munglani R, Schmidt WK, editors: *Pain: current understanding, emerging therapies, and novel approaches to drug discovery*, New York, 2003, Dekker.
12. Mendell LM: Modifiability of spinal synapses, *Physiol Rev* 64:260-324, 1984.
13. Woolf CJ, Salter MW: Neuronal plasticity: increasing the gain in pain, *Science* 288:1765-1769, 2000.
14. Mathews KA: Neuropathic pain in dogs and cats: if only they could tell us if they hurt, *Vet Clin North Am Sm Anim Pract* 38(6):1368, 2008.
15. Lei Z, Ruan Y, Yang AN et al: NMDA receptor mediated dendritic plasticity in cortical cultures after oxygen-glucose deprivation, *Neurosci Letts* 407:224-229, 2006.
16. Aida S: The Challenge of Preemptive Analgesia. Pain. Clinical updates, *IASP* 13(2):1, 2005.
17. Wall PD: The prevention of post-operative pain, *Pain* 32:289-290, 1988.
18. Fox SM, Mellor DJ, Firth EC et al: Changes in plasma cortisol concentrations before, during and after analgesia, anaesthesia and anaesthesia plus ovariohysterectomy in bitches, *Res Vet Sci* 57:110-118, 1994.
19. Hansen BD, Hardie EM, Carroll GS: Physiological measurements after ovariohysterectomy in dogs: what's normal? *Appl Anim Behav Sci* 51:101-109, 1997.
20. Sun WZ, Shyu BC, Shieh JY: Nitrous oxide or halothane, or both, fail to suppress *c-fos* expression in rat spinal cord dorsal horn neurons after subcutaneous formalin, *Br J Anaesth* 76:99-105, 1996.
21. Shafford HL, Lascelles BDX, Hellyer PW: Preemptive analgesia: managing pain before it begins, *Vet Med* 6:478-491, 2001.

22. Tverskoy M, Oz Y, Isakson A et al: Preemptive effect of fentanyl and ketamine on postoperative pain and wound hyperalgesia, *Anesth Analg* 78:205-209, 1994.
23. Rockemann MG, Selling W, Bischof C et al: Prophylactic use of epidural mepivacaine/morphine, systemic diclofenac, and methimazole reduces postoperative morphine consumption after major abdominal surgery, *Anesthesiology* 84:1027-1034, 1996.
24. Shaffran N: Pain management: the veterinary technician's perspective, *Vet Clin North Am Sm Anim Pract* 38(6):1423-1424, 2008.
25. Lee VC, Rowlingson JC: Pre-emptive analgesia: update on nonsteroidal anti-inflammatory drugs in anesthesia, *Adv Anesthes* V12:69-110, 1995.
26. Souter AJ, Fredman B, White PF: Controversies in the perioperative use of nonsteroidal anti-inflammatory drugs, *Anesth Analg* 79:1178-1190, 1994.
27. Gillies GWA, Kenny GNC, Bullingham RES et al: The morphine sparing effects of ketorolac tromethamine: a study of a new parenteral non-steroidal anti-inflammatory agent after abdominal surgery, *Anaesthesia* 42:727-731, 1987.
28. Sevarino FB, Sinatra RS, Paige D et al: The efficacy of intramuscular ketorolac in combination with intravenous PCA morphine for post-operative pain relief, *J Clin Anaesth* 4:285-288, 1992.
29. Hellyer P, Rodan I, Brunt J et al: AAHA/AAFP Pain Management Guidelines for Dogs & Cats, *J Am Anim Hosp Assoc* 43:235-248, 2007.
30. World Health Organization: *Cancer pain relief*, Geneva, 1986, World Health Organization.
31. Vernier VG et al: The toxicologic and pharmacologic properties of amantadine hydrochloride, *Toxicol Appl Pharmacol* 15:642-665, 1969.
32. Lascelles BDX, Gaynor JS et al: Amantadine in a multimodal analgesic regimen for alleviation of refractory osteoarthritis pain in dogs, *J Vet Intern Med* 22:53-59, 2008.
33. Cashmore RG, Harcourt-Brown TR et al: Clinical diagnosis and treatment of suspected neuropathic pain in three dogs, *Aust Vet J* 87:45-50, 2009.
34. Giori M, Saccomanni G et al: Pharmacokinetic evaluation of tramadol and its major metabolites after single oral sustained tablet administration in the dog: a pilot study, *Vet J* 180:253-255, 2009.
35. ERD®, Heska Corp.
36. Samad TA, Moore KA, Sapirstein A et al: Interleukin-1 beta-mediated induction of COX-2 in the CNS contributes to inflammatory pain hypersensitivity, *Nature* 401:471-475, 2001.
37. Wooten JG, Blikslager AT, Ryan KA et al: Cyclooxygenase expression and prostanoid production in pyloric and duodenal mucosae in dogs after administration of nonsteroidal anti-inflammatory drugs, *Am J Vet Res* 69:457-464, 2008.
38. Punke JP, Speas AL, Reynolds LR et al: Effects of firocoxib, tepoxalin and meloxicam on prostanoid and leukotriene production by duodenal mucosa and other tissues of osteoarthritic dogs, *Am J Vet Res* 69:1203-1209, 2008.
39. Omoigui S: The biochemical origin of pain—proposing a new law of pain: the origin of all pain is inflammation and the inflammatory response. Part 1 of 3—a unifying law of pain, *Med Hypoth* 69:70-82, 2007.
40. Kukanich B, Papich MG: Pharmacokinetics of tramadol and the metabolite O-desmethyltramadol in dogs, *J Vet Pharmacol Ther* 27:239-246, 2004.
41. Gottrup H, Juhl G, Kristensen AD et al: Chronic oral gabapentin reduces elements of central sensitization in human experimental hyperalgesia, *Anesthesiology* 101:1400-1408, 2004.
42. Curros-Criado MM, Herrero JF: The antinociceptive effect of systemic gabapentin is related to the type of sensitization-induced hyperalgesia, *J Neuroinflam* 4:15, 2007.
43. Fernihough J, Gentry C et al: Pain related behaviour in two models of osteoarthritis in the rat knee, *Pain* 112:83-93, 2004.
44. Kong VKF, Irwin MG: Gabapentin: a multimodal perioperative drug? *Br J Anaesth* 99:775-786, 2007.
45. Woolf CJ, Thompson SW: The induction and maintenance of central sensitization is dependent on N-methyl-D-aspartic acid receptor activation: implications for the treatment of post-injury pain hypersensitivity states, *Pain* 44:293-299, 1991.
46. Blanpied TA, Clarke RJ, Johnson JW: Amantadine inhibits NMDA receptors by accelerating channel closure during channel block, *J Neurosci* 25:3312-3322, 2005.
47. Nelson KA, Park KM, Robinovitz E et al: High-dose oral dextromethorphan versus placebo in painful diabetic neuropathy and postherpetic neuralgia, *Neurology* 48:1212-1218, 1997.
48. Sato J, Perl ER: Adrenergic excitation of cutaneous pain receptors induced by peripheral nerve injury, *Science* 251:1608-1611, 1991.
49. Pancrazio JJ, Kamatchi GL, Roscoe AK et al: Inhibition of neuronal Na+ channels by antidepressant drugs, *J Pharmacol Exp Ther* 284:208-214, 1998.
50. Jadad AR, Carroll D, Glynn CJ et al: Morphine responsiveness of chronic pain: double-blind randomized crossover study with patient controlled analgesia, *Lancet* 339:1367-1371, 1992.
51. Kukanich B, Lascelles BD, Aman AM et al: The effects of inhibiting cytochrome P450 3A, p-glycoprotein, and gastric acid secretion on the oral bioavailability of methadone in dogs, *J Vet Pharmacol Ther* 28:461-466, 2005.
52. Kukanich B, Lascelles BD, Papich MG: Pharmacokinetics of morphine and plasma concentrations of morphine-6-glucuronide following morphine administration to dogs, *J Vet Pharmacol Ther* 28:371-376, 2005.
53. Hansen B: How to prevent and relieve patient pain, *Vet Forum* 8:34-39, 1996.
54. Barnhart MD, Hubbell JAE, Muir WW et al: Pharmacokinetics, pharmacodynamics, and analgesic effects of morphine after rectal, intramuscular, and intravenous administration in dogs, *Am J Vet Res* 61:24-28, 2000.
55. World Health Organization: *WHO traditional medicine strategy 2002-2005*. Geneva, 2002, World Health Organization.
56. Gaynor JS: Acupuncture for management of pain, *Vet Clin North Am Small Anim Pract* 30:875-884, 2000.
57. Ma Y et al, editors: *Biomedical acupuncture for pain management: an integrative approach*, St. Louis, 2005, Elsevier.

58. Lin J-G, Chen W-L: Acupuncture analgesia: a review of its mechanisms of actions, *Am J Chin Med* 36(4):635-645, 2008.
59. Millis DL, Francis D, Adamson C: Emerging modalities in veterinary rehabilitation, *Vet Clin North Am Sm Anim Pract* 35(6):1335-1355, 2005.
60. Steiss JE, Levine D: Physical agent modalities, *Vet Clin North Am Sm Anim Pract* 35(6):1317-1333, 2005.
61. Canapp DA: Select modalities, *Clin Tech Small Anim Pract* 22:160-165, 2007.
62. Pai YC, Rymer WZ, Chang RW et al: Effect of age and osteoarthritis on knee proprioception, *Arthritis Rheum* 40:2260-2265, 1997.
63. van der Esch M, Steultjens M, Harlaar J et al: Joint proprioception, muscle strength, and functional ability in patients with osteoarthritis of the knee, *Arthritis Rheum* 57(5):787-793, 2007.
64. Burkholder WJ, Taylor L, Hulse DA: *Weight loss to optimal body condition increases ground reactive forces in dogs with osteoarthritis (abstract).* Purina Nutrition Forum Proceedings, St. Louis, 2000.
65. Impellizeri JA, Tetrick MA, Muir P: Effect of weight reduction on clinical signs of lameness in dogs with hip osteoarthritis, *J Am Vet Med Assoc* 216:1089-1091, 2000.
66. Canine osteoarthritis and omega-3 fatty acids. (Omega-3 fatty acids in canine osteoarthritis: A randomized, double-masked, practice based study. Effects of feeding omega-3 fatty acids on force plate gait analysis in dogs with osteoarthritis.) Hill's Clinical Evidence Report TD-827K.
67. Canine j/d clinical studies. (Dose titration effects of omega-3 fatty acids fed to dogs with osteoarthritis; A multi-center practice-based study of a therapeutic food and a nonsteroidal anti-inflammatory drug in dogs with osteoarthritis.) Hill's Clinical Evidence Report TD-828.
68. Budsberg S, Bartges J, Schoenherr W et al: *Effects of different n-6:n-3 fatty acid diets on canine stifle osteoarthritis* (abstract). Veterinary Orthopedic Society Annual Conference Proceedings, Lake Louise, Canada, 2001.

BIBLIOGRAPHY

1. Fox SM: *Chronic pain in small animal medicine*, London, 2009, Manson.
2. Fox SM, Millis DL: *Multimodal management of canine osteoarthritis*, London, 2009, Manson.
3. Mathews KA, ed: *Vet Clin North Am Sm Anim Pract* 38(6), 2008.
4. Gaynor JS, Muir WW, eds: *Handbook of veterinary pain management*, ed 2, St. Louis, 2009, Mosby Elsevier.
5. Muir WW, Woolf CJ: Mechanisms of pain and their therapeutic implications, *JAVMA* 219(10):1346-1356, 2001.
6. Kehlet H, Woolf CJ: Persistent postsurgical pain: risk factors and prevention, *Lancet* 367:1618-1625, 2006.
7. Mogil JS: Animal models of pain: progress and challenges, *Neuroscience* 283-294, 2009.
8. Jensen TS: Pathophysiology of pain: from theory to clinical evidence, *Eur J Pain Suppl* 2:13-17, 2008.
9. Lascelles BDX, Marcellin-Little DJ: *Practical approach to pain management and rehabilitation in canine osteoarthritis.* Proceedings of the Chronic Diseases Symposium, North American Veterinary Conference, Orlando, 2006.

15 The Role of Chondroprotectants, Nutraceuticals, and Nutrition in Rehabilitation

D.M. Raditic and J.W. Bartges

Veterinary rehabilitation is a rapidly developing field and is an integral therapy for many medically and surgically managed musculoskeletal, neurologic, and other conditions in dogs and cats. Nutrition plays an integral role in prevention and management of many conditions commonly seen in this field. This chapter discusses how nutrition can be used to help facilitate and promote healing and influence the rehabilitation process. Nutritional concepts such as overall caloric, protein, carbohydrate, vitamin, and mineral requirements are discussed, and a variety of supplements and their evidence are reviewed.

Nutritional Plan

To integrate appropriate nutritional recommendations for patient management, a two-step process is undertaken: a nutritional assessment phase (based on the Circle of Nutrition by the American College of Veterinary Nutrition; http://www.acvn.org) and an analysis, interpretation, and action phase.[1] In the assessment phase three sets of factors are assessed: animal-specific factors, diet-specific factors, and feeding management and environmental factors.[2] Animal-specific factors include age, physiologic status, disease processes, and activity of the patient. Diet-specific factors include safety and appropriateness of diet fed and nutrients of interest. Feeding factors include frequency, timing, location, and method of feeding, and environmental factors include living space and quality of the patient's surroundings.[1] An important part of the assessment phase is obtaining body weight, body condition score, and muscle condition score.[1] Body condition score evaluates body fat (Figure 15-1).[3-5] The goal for most patients is a body condition of 2.5 to 3.0 out of 5 or a 4 to 5 out of 9, which is associated with decreased health problems, including musculoskeletal conditions.[6-10] Muscle condition score evaluates muscle mass, which can be independent from body fat content assessed by body condition score (Figure 15-2).[11] Evaluation of muscle mass includes visual examination and palpation over temporal bones, scapulae, lumbar vertebrae, and pelvic bones. Muscle loss adversely affects strength, immune function, and wound healing. After collecting information from the assessment phase information is analyzed and interpreted and an action plan is implemented. After implementation of the action plan the patient undergoes repeated assessment and adjustment of the plan.

Role of Nutrition in the Prevention of Musculoskeletal Disease

Nutrition interacts with musculoskeletal disease. The prevalence of musculoskeletal disorders for all dogs at multicenter referral practices has been reported to be approximately one in four.[12] In dogs less than 1 year of age the prevalence is 22%, with 20% of these possibly having a nutrition-related cause.[13] We focus on two specific common conditions.

Developmental Orthopedic Disease

The role of nutrition in the development of musculoskeletal disease in growing dogs has been recognized for decades. Developmental orthopedic disease refers to a group of skeletal abnormalities (e.g., osteochondritis dissecans, hip dysplasia, wobblers' syndrome, ligamentous laxity, and hypertrophic osteodystrophy) that affect primarily rapidly growing, large- and giant-breed dogs. Nutrient excess (calcium and energy) and rapid growth (overfeeding and excess energy) are known risk factors for developmental orthopedic disease in dogs that have genetic risk.[14-18] Restricting food intake during growth slows growth rate without significantly reducing adult body size and is associated with decreased incidence of developmental orthopedic disease.[8,19] Dietary calcium greater than 3% on a dry matter basis is associated with increased risk despite an appropriate calcium-to-phosphorus ratio.[15] Even if the diet contains less calcium than this, excess calcium intake can occur if owners provide supplemental calcium. For example, two level teaspoons of calcium carbonate added to a growth diet fed to a 15-week-old Rottweiler puppy doubles the calcium intake. For large- and giant-breed dogs during growth, recommended dietary composition includes the following[20]:

- Energy: 3.2 to 4.1 kcal/kg of diet (lower end of range if at risk for developmental orthopedic disease or if clients use free-choice feeding)

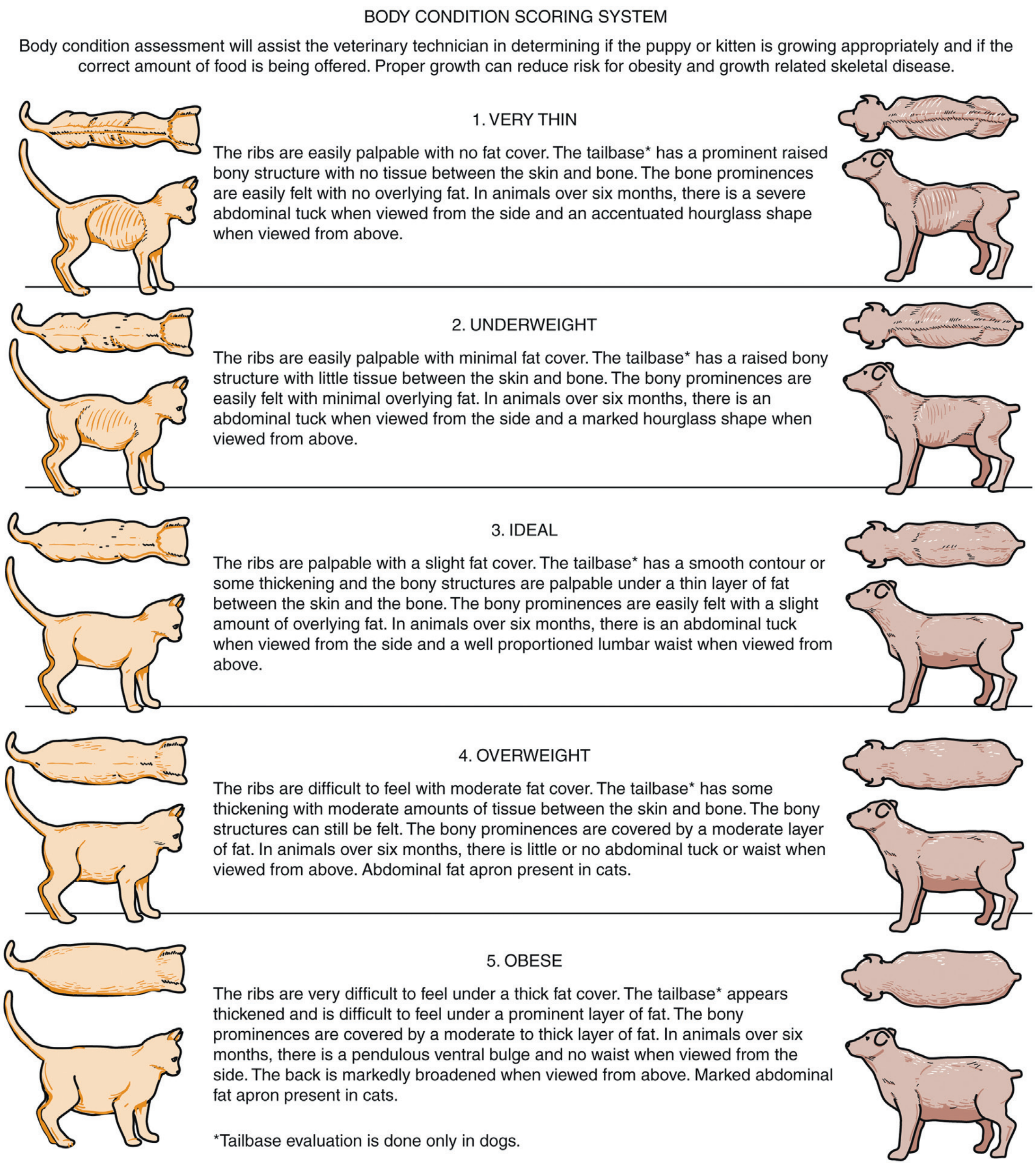
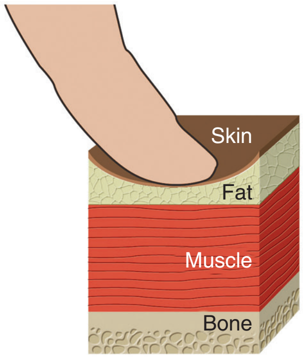


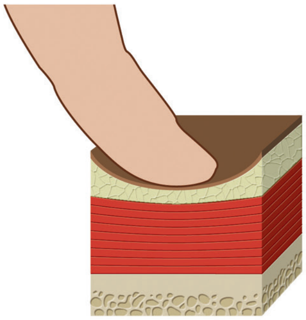
Figure 15-1 Body condition scoring (BCS) systems for the dog and the cat. (From Bassert JM, McCurnin DM: *McCurnin's clinical textbook for veterinary technicians,* ed 7. St. Louis, Saunders, 2010.)

Description **Figure**

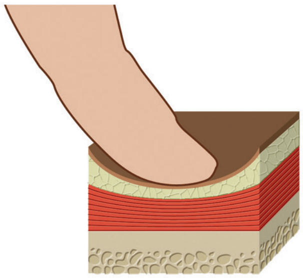
No muscle wasting, normal muscle mass



Mild muscle wasting



Moderate muscle wasting



Marked muscle wasting

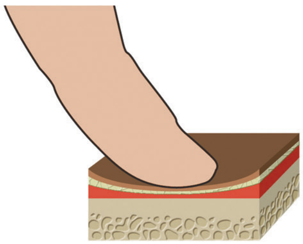


Figure 15-2 A muscle condition scoring (MCS) system. Evaluation of muscle mass includes visual examination and palpation over the temporal bones, scapulae, ribs, lumbar vertebrae, and pelvic bones. (Adapted from Baldwin K, Bartges J, Buffington T et al.: AAHA Nutritional assessment guidelines for dogs and cats, J Am AnimHospAssoc 46:285, 2010. IN Little S: The Cat: Clinical Medicine and Management, ed. 1, St. Louis, 2012, Saunders.)

- Crude fat: 8.5% to 11% on a dry matter basis
- Docosahexaenoic acid: ≥0.02% on a dry matter basis
- Protein: ≥27% on a dry matter basis
- Calcium: 0.8% to 1.2% (≤3% on a dry matter basis) with a calcium-to-phosphorus ratio of 1.1 to 2.0 : 1.0

The risk of developmental orthopedic disease is reduced if clients use limited meal-feeding rather than free-choice feeding and if growth rate is limited during the postweaning period.[21]

Obesity

Obesity can be defined as accumulation of body fat in excess of what is necessary to maintain optimum condition and health. This may be obvious in a grossly obese individual particularly when obesity-related disease is present; however, it is less obvious in a patient who is overweight and otherwise clinically healthy, but who is at risk for obesity-related disease such as osteoarthritis. Quantitatively, obesity is generally defined as exceeding ideal body weight by 15% to 20% or more. One technique that is useful in the management of clinical patients is a body condition score and muscle condition score as previously discussed. More sophisticated methods of body composition assessment include body composition determination by dual-energy x-ray absorptiometry (DEXA), quantitative computed tomography (quantitative CT) scans, or bioelectrical impendence.

Obesity may contribute to the development and progression of osteoarthritis because of excess forces placed on joints and articular cartilage, which may lead to inactivity and further development of obesity. Thus a vicious cycle ensues. Additionally adipose tissue is recognized as being metabolically active and proinflammatory; therefore obesity may contribute to inflammation.[22-26]

Several studies have demonstrated a relationship between obesity and osteoarthritis[9]; however, a cause and effect has not been determined.[27,28] A long-term study was performed of 48 Labrador retrievers from seven litters divided into two dietary groups. One group was fed an adult maintenance dog food at 0.27 kJ of metabolizable energy per kilogram of body weight per day, and the second group was fed the same diet at 75% of the amount (0.20 kJ of metabolizable energy per kilogram of body weight per day).[8,29] Restricted fed dogs lived on average 1.8 years longer, weighed less, had better body condition scores, and had longer delay to treatment of chronic disease including osteoarthritis.[21] In addition the restricted fed group had a lower incidence of osteoarthritis in most major joints, and if it occurred, it was less severe. Maintaining optimal or slightly lean body condition may be associated with lower risk of developing osteoarthritis, development of less severe osteoarthritis if osteoarthritis occurs, and delay of onset of clinical signs of osteoarthritis in dogs.

Role of Nutrition in the Treatment of Musculoskeletal Disease

Nutrition and Surgery

The role of nutrition in surgery in the context of physical rehabilitation and rehabilitation falls under two major categories: healthy surgical patients undergoing procedures for developmental (e.g., osteochondritis dissecans) or acquired (e.g., repair of cranial cruciate ligament damage) conditions, and posttrauma surgical patients undergoing procedures (e.g., fracture repair from motor vehicle injury).

Nontraumatized Surgical Patients

Patients undergoing surgery that are healthy and in optimal body and muscle condition (see Figures 15-1 and 15-2) require no further intervention nutritionally other than to continue feeding the current diet at the same amount and frequency. Patients are not fed 8 to 12 hours before anesthesia and surgery, but can resume normal feeding patterns once recovered from the procedure. It may be necessary to add flavoring agents or feed a different diet to stimulate appetite in the postoperative period. Patients should be assessed daily following surgery to insure they are eating adequately and that no adverse events occur either from the procedure or from postoperative medications such as analgesics and sedatives. Energy requirements during hospitalization and in the immediate postoperative period when activity is restricted is estimated in most patients to be the resting energy requirement (RER).[30,31]

$$\text{RER (kcal/day)} = 30\text{BW}_{\text{kg}} + 70$$

or

$$\text{RER (kcal/day)} = 70\text{BW}_{\text{kg}}{}^{0.75}$$

Although both equations are used to estimate RER, the authors prefer the second equation because body weight and RER are not linearly related as estimated with the first equation. As the patient becomes more active or as a rehabilitation program is initiated, energy requirements increase (see Nutrition and Rehabilitation).

In many instances surgery in these patients is elective; therefore, depending on the urgency of the surgical procedure, consideration should be given to weight reduction before surgery in patients that are obese (see Obesity). Reducing body weight to the optimal weight decreases anesthetic risk, improves immune function, and allows patients to more easily undergo postsurgical physical rehabilitation.[32]

Traumatized Surgical Patients

Patients with surgical musculoskeletal disease associated with trauma may or may not have increased nutritional requirements depending on the degree of trauma. If the trauma is less severe, then energy requirements are similar to nontraumatized surgical patients, and energy requirements approximate resting energy requirements.[30,33] Energy requirements are determined by estimating RER as stated.[30,31]

Severely injured patients or those with closed head injuries typically have energy requirements exceeding resting energy requirements because of the extent of the injuries or activation of the sympathetic nervous system, in part.[34] In these patients energy requirements may exceed two times the estimated resting energy requirements.[30,31]

Likewise protein requirements may exceed that required for maintenance to facilitate recovery, immune function, and wound healing.[35,36] Recommended protein intake for adult dogs at maintenance is 2.62 to 3.28 g/$\text{BW}_{\text{kg}}{}^{0.75}$ and for adult cats is 3.97 to 5.90 g/$\text{BW}_{\text{kg}}{}^{0.75}$.[37] With trauma or strenuous exercise, protein requirements may exceed these by 50% to 100%. Some patients may require facilitated nutritional support enterally or parenterally.[30,38-44] Nutritional support in these patients should be individualized and intensive monitoring is required to maximize recovery and response to supportive care. Convalescent diets are available that are formulated to contain higher levels of protein and fat are more calorically dense, highly palatable, and have a homogeneous texture suitable for feeding tubes.

Nutrition and Rehabilitation

Research into nutritional needs of patients undergoing various rehabilitation programs is nonexistent. Individuals developing rehabilitation programs should have knowledge of nutrition. Rehabilitation can include simple strengthening to extensive long-term physical rehabilitation; therefore, the nutritional plan for a patient must adjunctively meet the goals set during therapy and after goals are met. Nutritional plans should be included as part of the rehabilitation therapy. As described previously, a nutritional plan involves two phases: an assessment phase and an analysis, interpretation, and action phase. Reassessment is required until nutritional and rehabilitative goals are met.

Essential Nutritional Factors

Water

Water is the most essential nutrient. It is the transport medium for nutrients, heat, and waste products. Total body water is divided into four compartments with 64% being intracellular, 22% in interstitial spaces, 7% in plasma, and the remaining 7% transcellular.[45] Total body water is determined by total water intake and metabolic water production minus evaporation, urine, and fecal losses. An increase in muscle activity results in an increased demand for nutrients and removal of waste products. Only 20% to 30% of energy consumed by muscle tissue produces actual work, with the remaining 70% to 80% resulting in heat.[45]

Increased blood flow during exercise dissipates this heat. Increases in body heat during exercise must be removed because increases not only limit work, but also damage tissues.

During rehabilitation, excitement, fear, pain, and other stress factors affect the simplest of nutritional needs, water and electrolytes. These stress factors can increase body temperature and respiration alone or during exercise. It is important to monitor hydration and electrolyte balance during the program. A study of sprinting and endurance racing dogs showed water intake can directly affect biochemical and red blood cell profiles.[46] The same study showed increased blood cortisol and decreased blood glucose in sprint-racing sled dogs. These changes were attributed to metabolic stress and the physiologic response to high-intensity exercise.[46] High-intensity exercise is not commonly used in a rehabilitation setting, but the summation of patient stress, new exercise level, and perhaps changes in water and food intake should be noted.

Patients boarded for continuous rehabilitation may have poor diet and water intake in an unfamiliar environment.[45] Water and food intake should be recorded for hospitalized patients. Owners of outpatient dogs should be questioned about food intake, water intake, and other nutritional changes they may have noted during the program. Water intake for healthy dogs approximates 50 to 60 mL/kg per day, but may be more with rehabilitation.

Energy

Energy is a primary nutrient for maintenance, stage of life, and the therapy program. Diet supplies energy from carbohydrate, fat, and protein. Fats and carbohydrates are preferred substrates for exercise, whereas protein provides structural and functional tissue. Energy requirements correlate to work intensity, duration, and frequency, with intensity being the primary determinant. Studies show that sprinters depend on carbohydrates, whereas endurance activities use fat stores. Sprinters such as greyhounds that are running with maximum intensity have low energy requirements because of the short race duration and low frequency of training.[47] These dogs mainly use carbohydrates to meet energy requirements. Studies of endurance sled dogs with long durations of work are at the other extreme, requiring high energy intake. High-fat diets are fed to meet these high-energy requirements.[45,47,48]

Exercise intensity has not been quantified in rehabilitation, unlike performance and working dogs. Therapy is likely an intermediate level of exercise, which has not been well defined. Intensity, duration, and frequency of exercise are highly variable; therefore it is difficult to predict the energy needs of these patients. Consideration of historical exercise level combined with current patient and diet assessment may provide some insight into energy needs. A house pet with a body condition score (BCS) of 4 to 6/9 (see Figure 15-1), which usually has low energy requirements, that begins a rigorous short duration, increasing intensity, daily program, may need energy intake adjustment during rehabilitation.

The dog's daily energy requirement (DER) is based on the resting energy requirement (RER) multiplied by a factor for life stage or disease. Resting energy requirements are estimated as described in the preceding equations and defined as the energy requirement (kcal/day; 1 kcal = 4.184 kJ) for a healthy but fed animal at rest in a thermoneutral environment.[31] Maintenance, exercise, and life stage (growth, gestation, and lactation) are combined to develop factors of RER to estimate a patient's DER. Life stage factors commonly used in clinical nutrition are as follows[49]:

- Intact adult: 1.6
- Neutered adult: 1.4
- Obese prone: 1.2
- Weight loss: 1.0
- Gestation: 1.8 to 3.0
- Lactation: 4.0 to 8.0
- Growth: 1.6 to 2.5
- Adult, light work: 2.0
- Adult, moderate work: 3.0
- Adult, heavy work: 4.0 to 8.0
- Critical care: 1.0

These life stage factors are guidelines. Variables such as breed, disease, environment, and age can greatly affect energy requirements.

In rehabilitation, energy required to participate in a therapy program affects DER. If the therapist defines the treatment plan as work, then intensity, duration, and frequency of exercises are factored into the DER. Studies show sprinting greyhounds can have a DER of 1.6 to 2 × RER, whereas endurance sled dogs may require greater than 5 × RER.[47] This comparison is useful because there are nutritional studies of racing greyhounds that may apply to a rehabilitation program. Assigning an exercise factor to common physical rehabilitation programs would be useful in calculating the dog's DER. Until such data are provided it is important to recall that most house dogs have less than light exercise and may be obese or obese prone; therefore it is prudent to use low end exercise factors initially.

Reassessment of the patient during rehabilitation will determine if changes in energy intake are needed. If a patient loses weight during therapy, energy intake should increase. If weight gain is noted, energy intake should decrease. When starting a rehabilitation program with a patient with a BCS of 6 or 7, it may be best to make no energy changes despite the need for weight loss. The exercise program itself may induce weight loss and an ideal body condition score. Once the program is complete, client education about the patient's energy needs are discussed to maintain optimal body condition.

Fat

Fat provides the most calories in dog foods at 8.5 kcal/g of dry matter diet. Carbohydrates and protein supply 3.5 kcal/g of dry matter in a diet. To increase the energy density of a commercial diet, the fat content of the diet is increased. Fat also increases palatability of diets. Studies of endurance sled dogs show a positive effect for higher fat diets regarding performance. Sled dogs have been known to consume 80% of their caloric needs from fat.[45] In a study of well-conditioned beagles running on a treadmill, time to exhaustion correlated with fat intake, diet digestibility, and energy concentration of the diet.[50] Increasing fat intake improves performance of dogs with extreme energy requirements. In rehabilitation therapy, the level of exercise is unlikely to equate to the extreme energy requirements of sled dogs. Fat intake still should be considered when a patient is in therapy, however. If we use the example of a sprinting greyhound, dogs in therapy may have similar fat requirements with fat providing 20% to 30% of the total energy requirement. This equates to a diet with a fat content of 8% to 10% on a dry matter basis.[45] Stress factors of a new environment, exercise, fatigue, surgery, and illness may decrease diet intake. If there is appetite suppression, a higher-fat diet, which has a higher energy concentration and increased palatability, can be used to increase voluntary intake. Some dogs losing weight during a program are on diets too low in fat and energy concentration to meet the energy requirements of the rehabilitation program. The diet can be changed to a higher-fat diet with increased energy concentration to meet energy requirements. The type of fat in the diet should be evaluated. Diets must contain essential fatty acids in the form of unsaturated fatty acids. Linoleic acid, a 20-carbon n-6 fatty acid, is required in a dog's diet. N-3 fatty acid utilization is an area of extensive research in both humans and dogs and is discussed in a later section.

Carbohydrates

Carbohydrates are required only during gestation because dogs readily maintain normal blood glucose and tissue glycogen on a no-carbohydrate diet.[45] However, dogs can readily use carbohydrates for energy and most pet foods contain 40% to 60% carbohydrates on a dry matter basis. Racing greyhounds depend on carbohydrates that are metabolized into muscle glycogen. Muscle glycogen is a readily available source of energy for these sprinters with utilization of 70% of muscle stores in a single race. In humans, rats, and horses, increasing muscle glycogen improves exercise performance.[51,52] It is unknown whether increased carbohydrate intake in dogs enhances exercise performance or increases muscle glycogen stores. Studies have mixed results evaluating carbohydrate loading in racing greyhounds.[45,53] Extrapolation from human data suggests that 50% to 70% of total energy intake should be fed to maintain muscle glycogen stores for sprinting.[45] This is because anaerobic metabolism of glucose and glycogen is the dominant energy pathway that produces energy for sprinting. Because of the intensity of work and immediate energy needs for the racing greyhound, this extrapolation is logical, and these dogs are generally fed carbohydrates at a level that meets 50% or greater of the energy requirements. In physical rehabilitation, exercises are unlikely to have the intensity of work that is necessary to have an absolute carbohydrate requirement to employ anaerobic metabolism pathways. Carbohydrate intake using commercial dog foods appears reasonable.

Protein

Protein is required for structure and function, and to a lesser extent, energy. Current recommendations with exercise are to increase protein 5% to 15% regardless of the level of exercise, with dietary protein providing approximately 25% of the energy needs. Dietary protein is the most important nutrient with regard to muscle protein, and exercise increases protein requirements. Many of the goals of canine physical rehabilitation require increased muscle tissue. Studies of protein intake during exercise are needed to identify ideal protein intake for dogs during physical rehabilitation.

In a postabsorptive state, protein that has been broken down to amino acids provides the substrate for protein synthesis elsewhere in the body. Exercise increases protein synthesis and breakdown with the net result of decreased muscle protein. Ingestion of amino acids stimulates muscle protein synthesis, exceeding the rate of breakdown, and muscle mass increases. In one study of elderly humans, patients maintained on bed rest for 10 days were given a dietary supplement of amino acids or placebo in a double-blind randomized fashion.[54] Supplementing essential amino acids in these individuals with absolute bed rest prevented decreases in muscle strength and function. Furthermore amino acid supplementation prevented the decreased rate of muscle synthesis that is normally seen with bed rest.[54] Amino acid stimulation of muscle protein synthesis and exercise is synergistic. Exercise combined with amino acid intake produces a greater response regarding muscle synthesis than response to individual treatments. This suggests exercise can prime the muscle response to the anabolic effects of amino acids or exercise activates muscle synthesis, but adequate amino acid precursors are needed to increase muscle mass. Postexercise supplementation with carbohydrate and protein or protein hydrolysate increases muscle glycogen stores and may facilitate recovery.[55-57] The effect of amino acid supplementation via infusion on muscle proteolysis in beagles was found to require concomitant glucose infusion; therefore an energy substrate for protein anabolism is required for the desired beneficial effect.[58] Branched chain amino acids (leucine, isoleucine, and valine) supplementation decreases branched chain amino acid catabolism and exercise-induced muscle damage promoting muscle protein synthesis in humans[59-61]; however, results are contradictory and branched chain amino acids may only facilitate recovery of muscle following exercise.[62] Alanine supplementation improves high-intensity aerobic performance of muscle.[63] Carnitine is a quaternary ammonium compound biosynthesized from lysine and methionine and is a cofactor for transport of long-chain fatty acids across the mitochondrial membrane to liberate energy via β oxidation. In dogs, it has been shown to improve function and energy metabolism in muscle and may be useful in rehabilitation.[64,65] However glutamine supplementation does not appear to be beneficial in preventing muscle protein catabolism with exercise.[66]

Studies of protein nutrition often evaluate total dietary protein content instead of amino acid composition. Investigation to optimize amino acid profiles for specific benefits has identified "slow" and "fast" proteins. This classification is determined by the rate a protein exits the stomach, is hydrolyzed into peptides and amino acids, and is then absorbed. Plasma amino acid profiles of fast and slow proteins have shown that "blends" are more likely to produce maximal amino acid concentration for optimal protein synthesis. The blends maintain amino acid concentrations for an optimal time for longer duration of anabolism. Identifying and using protein blends for maximal muscle synthesis and strength during physical rehabilitation may be a reasonable goal with these current studies.

Obesity

Weight reduction from an obese state is beneficial in the management of osteoarthritis in humans.[67-69] In addition to improvement in mobility, weight reduction was associated with better quality of life scores in humans with knee osteoarthritis.[70] Three uncontrolled clinical studies of obese dogs with osteoarthritis demonstrated improvement in mobility.[71,72] In one study of nine dogs with body condition score of 5 out of 5 and coxofemoral osteoarthritis, obesity management resulted in loss of 11% to 18% of body weight, a decrease in body condition to optimal, and improvement in severity of subjective hind limb lameness scores.[71] In the second study of 16 dogs with coxofemoral osteoarthritis, weight loss of 13% to 29% of body weight and decrease of body condition to optimal resulted in improvement of ground reactive forces as well as improvement in subjective mobility and clinical signs of osteoarthritis.[72] In the third study, 14 client-owned dogs with clinical and radiographic signs of osteoarthritis participated in an open prospective clinical trial. Results indicated that body weight reduction caused a significant decrease in lameness with a weight loss of 6.10% or more. Kinetic gait analysis supported the results with a body weight reduction of 8.85% or more. These results confirm that weight loss should be presented as an important treatment modality to owners of obese dogs with osteoarthritis, and that noticeable improvement may be seen after modest weight loss of 6.10% to 8.85% body weight. Although these are uncontrolled clinical trials of a small number of dogs, results suggest that weight reduction may be beneficial in dogs with osteoarthritis.

Weight loss is beneficial when used in combination with physical rehabilitation in dogs. In a non-blinded prospective randomized clinical trial, 29 adult dogs that were overweight or obese (body condition score of 4/5 or 5/5) having clinical and radiographic signs of osteoarthritis were evaluated.[73] All dogs underwent weight loss. One group received caloric restriction and a home-based physical rehabilitation program and the other group received an identical dietetic protocol and an intensive physical rehabilitation program, including transcutaneous electrical nerve stimulation. Significant weight loss and improved mobility were achieved in both groups; however, greater weight reduction and better mobility was obtained in the group receiving intensive physical rehabilitation.[73]

Nutritional Compounds Evaluated in Dogs or In Vitro Using Canine Cells

There are other dietary supplements recommended for patients with osteoarthritis.[74-87] Very few have been evaluated in a controlled manner and even fewer have been evaluated in dogs with osteoarthritis.

Antioxidants

Formation of free radicals as a consequence of cellular metabolism occurs constantly, but potential deleterious effects are minimized by antioxidants. Balance between free radicals and antioxidant defenses is an essential factor in preventing development of noxious processes at cellular and tissue levels. Recent evidence supports the theory that excessive production of free radicals or the imbalance between concentrations of free radicals and antioxidant defenses may be related to processes such as aging, cancer, diabetes mellitus, lupus, and arthritis.[88] Progressive hypoxia resulting in production of reactive oxygen species may play a role in rheumatoid arthritis.[89] Thus antioxidant therapy may be of benefit in the treatment of osteoarthritis.[90]

S-Adenosylmethionine. S-adenosylmethionine (SAMe) is a co-substrate involved in transmethylation, transsulfuration, and aminopropylation reactions, which occur primarily in the liver. In controlled trials of humans with osteoarthritis, SAMe was as effective as nonsteroidal anti-inflammatory drugs (NSAIDs) and better than placebo in reducing pain and improving function with a lower likelihood of side effects[91-98]; however, no difference with an NSAID drug was found in one study.[99] A systematic review was inconclusive and evidence of efficacy was hampered by inclusion of small trials of questionable quality.[100] A double-blinded, placebo-controlled clinical trial of 33 dogs with presumed osteoarthritis did not show a benefit of

SAMe as a sole treatment.[101] An in vitro study showed SAMe adversely affected chondrocyte viability.[102] SAMe may reduce inflammatory mediators, increase levels of the antioxidant glutathione, increase cartilage synthesis, be chondroprotective, and maintain DNA methylation.[103]

Other Antioxidants. Most studies of antioxidant therapy for arthritis in humans involve rheumatoid arthritis, although several studies of knee osteoarthritis have been published. An early pilot study of vitamin E (tocopherols) demonstrated improvement in clinical signs and pain scores in 52% of 29 patients with knee osteoarthritis compared with 4% receiving placebo.[104] In another placebo-controlled study of humans with osteoarthritis of coxofemoral or knee joints, high-dose tocopherols was as efficacious as diclofenac in reducing pain and improving mobility.[105] This was also found in studies of humans with rheumatoid arthritis.[106,107] Other studies have not demonstrated a benefit of vitamin E for either rheumatoid arthritis or osteoarthritis.[108-110] Vitamin E has not been evaluated in dogs with osteoarthritis; however, large breed puppies fed a diet proportionately higher in protein, calcium, n-3 fatty acids, and antioxidants had increased lean body mass and improved cartilage turnover as maturity was attained.[111] In greyhounds and sled dogs, supplementation is associated with increased plasma levels; however, no difference in muscle damage with exercise has been noted.[112-114] Efficacy of vitamin C in humans with osteoarthritis has been difficult to determine due to contradictory results.[115-118] In dogs, vitamin C administration was found to increase vitamin C plasma levels, but was associated with decreased racing performance in greyhounds.[119] Selenium was not shown to be beneficial in humans with rheumatoid arthritis.[120] An antioxidant cocktail was administered for 6 weeks to 48 dogs. Dogs were assigned to four groups: untrained/not supplemented, untrained/supplemented, trained/not supplemented, and trained/supplemented. Metabolomic profiling showed that dogs receiving the antioxidant cocktail recovered to baseline values 24 hours after exercise, whereas dogs receiving no supplementation did not. Therefore administration of an antioxidant cocktail apparently facilitates recovery from exercise.[121]

Avocado/Soybean Unsaponifiables

Avocado/soybean unsaponifiables (ASUs) are composed of the unsaponifiable fractions of avocado and soybean oils in a one-third to two-thirds proportion.[122,123] Avocado/soybean unsaponifiables have anti-osteoarthritic properties by inhibiting interlukin-1 and stimulating collagen synthesis in cartilage cultures.[123-125] In vivo there is one report in an ovine meniscectomy model of osteoarthritis that supports disease-modifying capabilities.[125,126] Human clinical trials have shown some beneficial effects of ASUs on clinical symptoms of osteoarthritis and they may have some structure modification capabilities.[127-129] However there are conflicting data as other studies found no long-term benefits.[128,130-132] In one study of dogs, osteoarthritis was induced by anterior cruciate ligament transection that then received placebo or ASUs (10 mg/kg per 24 hours).[133] The size of macroscopic lesions of the tibial plateau, severity of cartilage lesions, synovial cellular infiltration, and inducible nitric oxide synthase were decreased significantly and there was reduced loss of subchondral bone volume and calcified cartilage thickness in the group receiving ASUs.[133]

Blueberries

Blueberries are flowering plants of the genus *Vaccinium*, which includes cranberries and bilberries. They may have antioxidative and antiinflammatory actions because they contain pterostilbene, anthocyanins, proanthocyanidins, resveratrol, flavonols, and tannins.[134] In a study of sled dogs, those receiving a blueberry supplement had higher antioxidant status than those not receiving the supplement; however, there was no difference in creatinine kinase activity and isoprostane levels (a measure of oxidative stress) in dogs receiving blueberries and those that did not.[135]

Boswellia serrata

Boswellia, also known as Boswellin or Indian frankincense, comes from the Indian *Boswellia serrata* tree. Resin from the bark of this tree is purported to have antiinflammatory properties derived primarily from 3-O-acetyl-11-keto-β-boswellic acid (AKBA), which inhibits 5-lipoxygenase and matrix metalloproteinases, and decreases tumor necrosis factor α and interleukin 1β.[136,137] Following oral administration, AKBA has poor bioavailability. However, two AKBA-containing compounds, Alfapin and 5-Loxin, have good bioavailability.[138] Boswellia resin has been shown to improve clinical signs and pain in humans in controlled studies.[139-141] Boswellia resin has been evaluated in 24 dogs in an open multicenter study.[142] Improvement in clinical signs, lameness, and pain was found in 17 of 24 dogs. Diarrhea and flatulence occurred in five dogs.

Chondroprotectants

Chondromodulating agents are purported to slow or alter the progression of osteoarthritis. These agents are considered to be slow-acting drugs in osteoarthritis (SADOA) and can be subdivided into symptomatic slow-acting drugs (SYSADOA) and disease-modifying osteoarthritis drugs (DMOAD). Beneficial effects may include a positive effect on cartilage matrix synthesis and hyaluronan synthesis by synovial membrane, as well as an inhibitory effect on catabolic enzymes in osteoarthritis joints.[143] Compounds fall under two different categories. One group includes agents that are approved by the US Food and Drug Administration, such as polysulfated glycosaminoglycan, which can have label claims of clinical effects. The second group includes products that are considered to be nutritional supplements, which are not required to undergo efficacy testing, and legally cannot claim any medical benefits. Examples of this group include glucosamine and chondroitin sulfate. Although many of these products are administered as a supplement or alternative treatment, some, such as glucosamine and green-lipped mussels, are incorporated into pet foods.

Glucosamine and Chondroitin. Glucosamine is an amino sugar that is a precursor for biochemical synthesis of glycosylated proteins and lipids. D-glucosamine is made naturally in the form of glucosamine-6-phosphate and is the biochemical precursor of nitrogen-containing sugars.[144] Specifically glucosamine-6-phosphate is synthesized from fructose-6-phosphate and glutamine[145] as the first step of the hexosamine biosynthesis pathway. The end product of this pathway is UDP-N-acetylglucosamine, which is used for making glycosaminoglycans, proteoglycans, and glycolipids. Because glucosamine is a precursor for glycosaminoglycans, and glycosaminoglycans are a major component of joint cartilage, supplemental glucosamine may help enhance cartilage synthesis and there are in vitro and clinical data to support this claim, although in vitro studies often use concentrations not achieved in serum or plasma after oral administration.[146-152] There are conflicting data on evidence of clinical effect of glucosamine-chondroitin in osteoarthritis.[74,75,153-161] In a randomized, double-blind, positive-controlled clinical trial comparing glucosamine hydrochloride and chondroitin sulfate to carprofen in 35 dogs with osteoarthritis, carprofen-treated dogs had improvement in five subjective measures, whereas dogs treated with glucosamine-chondroitin sulfate had improvement in three of five measures, but only at the final assessment point of 70 days.[162] A 60-day prospective, randomized, double-blinded, placebo-controlled trial of 71 dogs with osteoarthritis assessed subjective and objective measures comparing carprofen, meloxicam, glucosamine-chondroitin, and placebo. Results indicated that objectively measured variables improved significantly with carprofen and meloxicam, but not with glucosamine-chondroitin or placebo. Subjective findings of veterinarians agreed with results of objective evaluation, but subjective assessment by owners identified improvement only with meloxicam.[163] Based on these results there is weak clinical evidence of benefit of glucosamine-chondroitin in dogs with osteoarthritis.[153-155]

When evaluating glucosamine and chondroitin sulfate inclusion in a manufactured dog food, two questions should be asked. Is the glucosamine and chondroitin sulfate in the food stable and bioavailable? And, is the amount of glucosamine and chondroitin sulfate in the food enough to provide a beneficial effect? It is difficult to find the amount of glucosamine and chondroitin sulfate in pet foods. Many dog foods formulated and marketed for adult dogs, geriatric dogs, and large breed growth contain glucosamine and chondroitin sulfate, but the amounts are not readily available from the manufacturer or from electronic or print information. These compounds are not recognized by the American Association of Feed Control Officials, and so are not included in dog nutrient profiles. Furthermore, they are not considered as generally regarded as safe (GRAS) ingredients. However, there is no evidence to suggest that they are not safe at the levels used in food.

The New Zealand Green-Lipped Mussel

The New Zealand green-lipped mussel (*Perna canaliculus*) is a rich source of glycosaminoglycans, although its proposed benefit is thought to be from its antiinflammatory effects.[164] In 1986 dried mussel extracts became available that were stabilized with a preservative. The earlier studies that found no beneficial effect of the green-lipped mussel on arthritis used preparations that had not been stabilized, a point that may help explain some of the discrepancies in the research. A stabilized lipid extract is more effective than a nonstabilized extract at inhibiting inflammation.[165] In an uncontrolled clinical trial stabilized lipid extract administration resulted in an 80% improvement in humans with rheumatoid arthritis.[166] In a randomized controlled clinical study of 31 dogs with arthritis, green-lipped mussel powder (0.3%) was added to the diet during processing for one group of dogs.[167] Compared with the control group, which was fed the same diet without added green-lipped mussel powder, there were significant improvements in subjective arthritis scores, joint swelling, and joint pain in the treated group. However, these data must not be overinterpreted. In a randomized, double-controlled and double-blinded clinical trial, 45 dogs with osteoarthritis were evaluated. Dogs receiving green-lipped mussels had improved mobility when compared with placebo; however, it was not as effective as carprofen.[168] In a study of 81 dogs with presumptive osteoarthritis, clinical signs improved in both the green-lipped mussel and placebo-treated groups when compared with baseline. On day 56 of the study, dogs receiving green-lipped muscle had improved clinical signs but not musculoskeletal scores.[169] An uncontrolled study of 85 dogs that were fed a green-lipped mussel supplemented diet for 50 days showed reduction of a composite arthritic score when compared with baseline scores on various diets dogs were consuming.[170] In systematic reviews of agents used to treat canine osteoarthritis, the data regarding the benefits of green-lipped mussel extract in dogs were promising but uncertainties existed relating to the scientific quality of the data and no definitive relationship has been proved between clinical improvements and the therapy.[154,155]

Omega-3 Fatty Acids

Degenerative osteoarthritis involves an inflammatory component. Thus, it may be possible to modify the inflammation by nutritional components, specifically omega-3 (n-3) fatty acids. Arachidonic acid (an n-6 fatty acid) is incorporated into cell membranes, and when metabolized, it yields prostaglandins, leukotrienes, and thromboxanes of the two and four series. Many drugs used to treat degenerative osteoarthritis inhibit conversion of arachidonic acid to these eicosanoids. These n-6-derived eicosanoids have, for the most part, vasoactive and proinflammatory effects. Substituting an n-3 fatty acid in the membrane may decrease these responses. Metabolism of n-3 fatty acids results in eicosanoids of the three and five series, which are less vasoactive and less proinflammatory. In addition to modulating cytokines, n-3 fatty acids have been shown to reduce expression of cyclooxygenase-2, lipoxygenase-5, aggrecanase, matrix metalloproteinase 3 and 13, interleukin-1α and β, and tumor necrosis factor α.[171-175] In addition to these effects, novel oxygenated products generated from n-3 fatty acids, eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) have been identified in resolving inflammatory exudates and tissues and are called Resolvins (resolution phase interaction products) and docosatrienes.[176-178] They may have a role in inflammatory diseases such as osteoarthritis.[179-184]

Several studies have demonstrated a beneficial response to n-3 fatty acid incorporation into diets of humans with rheumatoid arthritis,[185-189] although other studies have not demonstrated a benefit.[190] There is a growing body of data showing positive effects of n-3 fatty acids on cartilage and its metabolism in the face of degradative enzymes; n-3 fatty acid supplementation can reduce inflammatory and matrix degradative responses elicited by chondrocytes during osteoarthritis progression.[172-174,191] Eicosapentaenoic acid appears to be more effective than DHA and α-linolenic acid.[171]

An unpublished study was performed in dogs evaluating omega-3 fatty acids and experimentally induced stifle arthritis.[192,193] Eighteen dogs were randomly assigned to one of three isocaloric diet groups containing 21.4% fat (dry matter basis) differing only in their fatty acid composition: a diet with an n-6 to n-3 fatty acid ratio of 28.0 : 1.0 (high n-6 diet), a diet with an n-6 to n-3 fatty acid ratio of 8.7 : 1.0 (control diet), and a diet with an n-6 to n-3 fatty acid ratio of 0.7 : 1.0 (high n-3 diet). Diets were fed for a total of 21 months. Dogs were begun on their assigned diet 3 months before surgical transection of the left cranial cruciate ligament, then continued until surgical repair 6 months later, and were maintained for an additional 12 months following repair. When compared with the high n-6 and control diets, consumption of the high n-3 diet was associated with lower serum concentrations of cholesterol, triglycerides, and phospholipids, lower synovial concentration of prostaglandin E2, better ground reaction forces, and less radiographic changes of osteoarthritis. Synovial membrane fatty acid composition mirrored the fatty acid composition of the diets consumed by the dogs. Another study in dogs reported results of owner observations who perceived improvement in their pet's arthritic signs when treated with fatty acids for various dermatologic problems.[194] In a study of 127 dogs with osteoarthritis, dogs fed a diet with a 31-fold increase in total n-3 fatty acid content and a 34-fold decrease in n-6 to n-3 fatty acid ratio for 6 months had improved ability to rise from a resting position and play when compared with dogs fed a control diet.[195] In a randomized, double-blinded, controlled clinical trial of 38 dogs with osteoarthritis, dogs fed a diet containing 3.5% n-3 fatty acids for 90 days had improved peak vertical force values and subjective improvement in lameness and weight bearing when compared with a control diet.[196] Another randomized, controlled clinical trial was performed with 131 dogs with stable chronic osteoarthritis treated with carprofen. Over a 12-week study carprofen dosage decreased significantly faster in dogs fed a diet supplemented with n-3 fatty acids when compared with a control diet.[197] Based on results of these studies there is a rationale for n-3 fatty acid supplementation or feeding diets containing increased n-3 fatty acid levels to dogs with osteoarthritis.

P54FP (Turmeric)

P54FP is an extract of Indian and Javanese turmeric, *Curcuma domestica* and *Curcuma xanthorrhiza*, which contains a mixture of active ingredients including curcuminoids and essential oils.[198] There is evidence that these active ingredients possess antiinflammatory activity. Specifically curcumin has been shown to inhibit prostaglandin E2, cyclooxygenase-2, nuclear factor κB, matrix metalloproteinase-3 upregulation, and IL-1β–induced proteoglycan degradation.[199-201] One randomized, blinded study in dogs with osteoarthritis found no difference in objective ground reaction forces but did see some improvements in certain subjective outcome measures.[198]

Special Milk Protein Concentrate

Milk contains a number of biologically active compounds including immunoglobulins, cytokines, enzymes, hormones, and growth factors. These compounds impart antiinflammatory properties that have been recognized in human breast milk[202] and milk from hyperimmunized cows.[203,204] A special milk protein concentrate prepared from milk of hyperimmunized cows exerts antiinflammatory properties.[205] The antiinflammatory property does not appear to be caused by inhibition of arachidonic acid metabolism, but by suppression of neutrophil migration from the vascular space.[206] A randomized, controlled clinical trial has been performed evaluating special milk protein concentrate (Microlactin, Stolle Milk Biologics, Cincinnati, OH) in dogs with naturally occurring osteoarthritis.[207] In this study dogs receiving the special milk protein concentrate had improvement in subjective clinical signs of osteoarthritis and owner global assessment compared with dogs receiving placebo. Further studies need to be completed to validate these findings.

Zeel®

Zeel® is an over-the-counter homeopathic preparation based on highly diluted extracts from plants, animals, and minerals (sulfur) as well as defined biochemical substances including coenzyme A, DL-alpha lipoic acid, sodium diethyl oxalate, and nicotinamide adenine dinucleotide.[208,209] It has been evaluated in two controlled studies in dogs. Active ingredients (common name) include *Arnica montana* (mountain arnica), *Sanguinaria canadensis* (blood root), *Solanum dulcamara* (bittersweet), *Symphytum officinale* (comfrey), *Toxicodendron quercifolium* (poison oak), *Cartilago suis* (porcine cartilage), *Embryo totalis suis* (porcine embryo), *Funiculus umbilicalis suis* (porcine umbilical cord), *Placenta totalis suis* (porcine placenta), coenzyme A, dl-α-lipoic acid, nicotinamide adenine dinucleotide, sulfur, and sodium diethyl oxalate.[208,209] In one study in dogs (n = 68) greater than 1 year of age diagnosed with osteoarthritis, it was compared with carprofen in a multicenter, prospective, observational open-label cohort study in 12 German veterinary clinics.[208] In another study in dogs (n = 44) greater than 1 year of age diagnosed with osteoarthritis, it was compared with carprofen and a placebo.[209] Clinical signs and several measures of osteoarthritis improved significantly with treatment in both studies; however in one study,[209] it was not as effective as carprofen. The composition of the products and the dosage of Zeel® differed between the two studies, which confounds interpretation of results.

Other Nutritional Compounds Evaluated in Other Species or In Vitro

Articulin F

Articulin F is an herbo-mineral formulation containing roots of *Withania somnifera*, the stem of *Boswellia serratia*, rhizomes of *Curcuma longa*, and a zinc complex. In a double-blind, placebo-controlled study of 42 humans with osteoarthritis, Articulin F was associated with improvement in pain and function, but not in radiologic findings.[139]

Barberry

Extracts obtained from the roots and barks of various *Berberis* species are used as folk remedies worldwide for the treatment of various inflammatory ailments. Effects of the extracts and fractions from the roots of *Berberis crataegina* have been studied using various in vivo models of inflammation in mice and rats and have resulted in potent inhibitory activity against carrageenan- and serotonin-induced hind paw edema, acetic acid-induced increased vascular permeability, castor oil–induced diarrhea, and Freund's complete adjuvant–induced (FCA) arthritis models.[210,211]

Boron

Boron deficiency in food may be part of the cause of some arthritides.[212] Epidemiologic studies suggest that humans in countries with low boron intake (less than 1.0 mg/day) have a higher risk of arthritis development when compared with humans in countries in which boron intake is higher (3-10 mg/day).[213] In a double-blind, placebo versus boron supplementation trial with 20 subjects with osteoarthritis, a significant favorable response to a 6-mg boron/day supplement was found; 50% of subjects receiving supplement improved compared with 10% receiving the placebo.[213] Whether boron deficiency occurs in pet dogs is unknown. Furthermore whether boron supplementation and what dosage of boron would be best is unknown.

Cat's Claw

Cat's claw, an Amazonian medicinal plant (*Uncaria tomentosa* and *U. guyanensis*), has antiinflammatory and antioxidant effects and has been shown to decrease clinical signs of knee arthritis and rheumatoid arthritis in humans.[214,215] Cat's claw has been found to inhibit lipopolysaccharide-induced inducible nitric oxide gene expression, nitrate formation, cell death, PGE_2 production, activation of NF-κB, and TNFα.[216-219] It also acts as a free radical scavenger, suppresses TNFα and apoptosis, and is cytoprotective against oxidative damage.[217,220] It has been used to treat osteoarthritis in humans. In a metaanalysis of available data,[81] three studies demonstrated effectiveness of cat's claw alone or in combination with other treatment for osteoarthritis.

Cetyl Myristoleate

Cetyl myristoleate is the cetyl ester of myristoleic acid and has antiinflammatory and immune-modulating activity.[221] Despite conflicting results in rodent models,[222,223] it was shown to improve function and range of motion when compared with placebo in 64 humans with knee osteoarthritis.[224]

Chinese (Baikal) Skullcap

Chinese or Baikal skullcap (*Scutellaria baicalensis*) is a flowering plant used in traditional Chinese medicine.[81] Sometimes *Scutellaria lateriflora* (North American Skullcap) is mistaken for *Scutellaria baicalensis* (Baikal Skullcap). This can result in intake of the incorrect species that is often contaminated with other plants at high enough levels to be hepatotoxic.[225] Chinese skullcap or its B-ring flavones and flavonoids have been shown to exhibit antispasmogenic, antivasoconstrictor, antihypertensive, antiplatelet aggregation, and antiinflammatory properties including inhibition of cyclooxygenase in vitro.[81,226-229] Flavocoxid (Limbrel®, Primus Pharmaceuticals Inc., Scottsdale, AZ) contains a proprietary blend of free-B ring flavonoids and flavins from the root of *Scutellaria*